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Inventor(s)	Tanaka; Rui

Radio-frequency module and communication device

Abstract

A radio-frequency module includes a mounting substrate, a first filter, a second filter, a shield layer, and a conductor. The mounting substrate has a first main surface and a second main surface on opposite sides. The shield layer is disposed on an outer surface of a resin layer with which the first filter and the second filter are covered. The radio-frequency module is capable of performing simultaneous transmission by using both the first filter and the second filter. The conductor is disposed on the first main surface of the mounting substrate and is in contact with the transmitting filter and the mounting substrate. The conductor is in contact with the shield layer on a side other than a side closer to the second filter than to the first filter.

Inventors:	Tanaka; Rui (Kyoto, JP)
Applicant:	Murata Manufacturing Co., Ltd. (Kyoto, JP)
Family ID:	1000008778240
Assignee:	MURATA MANUFACTURING CO., LTD. (Kyoto, JP)
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Primary Examiner: Nguyen; Hoa C

Attorney, Agent or Firm: McDonald Hopkins LLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a continuation of International Application No. PCT/JP2021/040194 filed on Nov. 1, 2021 which claims priority from Japanese Patent Application No. 2020-183885 filed on Nov. 2, 2020. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND ART

Technical Field

(1) The present disclosure relates to a radio-frequency module and a communication device and, more specifically, to a radio-frequency module including a first filter and a second filter and to a communication device including the radio-frequency module.

(2) A radio-frequency power-amplifying module (radio-frequency module) including two transmitting filters (a first filter and a second filter) whose passbands are different communication bands is disclosed in Patent Document 1. The two transmitting filters of the radio-frequency power-amplifying module can work concurrently. Patent Document 1: International Publication No. 2016/117482

BRIEF SUMMARY

(3) When the radio-frequency power-amplifying module disclosed in Patent Document 1 performs simultaneous transmission by using both a first transmitting filter (first filter) and a second transmitting filter (second filter), heat generated by the first transmitting filter (first filter) affects the second transmitting filter (second filter). This can cause deterioration of the characteristics of the second transmitting filter.

(4) The present disclosure provides a radio-frequency module and a communication device that are capable of eliminating or reducing the possibility that heat generated by a first filter of the radio-frequency module will cause deterioration of the characteristics of a second filter of the radio-frequency module while both the first filter and the second filter are used for simultaneous transmission.

(5) According to an aspect of the present disclosure, a radio-frequency module includes a mounting substrate, a first filter, a second filter, a resin layer, a shield layer, and a conductor. The mounting substrate has a first main surface and a second main surface on opposite sides. The first filter is mounted on the first main surface of the mounting substrate and is a filter whose passband is a first frequency band. The second filter is mounted on the first main surface of the mounting substrate and is a filter whose passband is a second frequency band. The resin layer is disposed on the first main surface of the substrate. The first and second filters are at least partially covered with the resin layer. The shield layer is disposed on an outer surface of the resin layer. The conductor is disposed on the first main surface of the substrate and is in contact with the first filter and the substrate. The shield layer is disposed on the outer surface of the resin layer. The radio-frequency module is capable of performing simultaneous transmission by using both the first filter and the second filter. The conductor is in contact with the shield layer on a side other than a side closer to the second filter than to the first filter.

(6) According to an aspect of the present disclosure, a communication device includes the radio-frequency module according to the aforementioned aspect and a signal processing circuit. The signal processing circuit is connected to the radio-frequency module and is configured to process a radio-frequency signal.

(7) The radio-frequency module and the communication device according to the present disclosure are capable of eliminating or reducing the possibility that heat generated by the first filter of the radio-frequency module will cause deterioration of the characteristics of the second filter of the radio-frequency module while both the first filter and the second filter are used for simultaneous transmission.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of a radio-frequency module and a communication device according to an embodiment.
- (2) FIG. 2 is a plan view of the radio-frequency module.
- (3) FIG. 3 is a sectional view taken along line Z1-Z1 in FIG. 2.
- (4) FIG. 4 is a plan view and illustrates an arrangement of two transmitting filters capable of working concurrently and conductors provided for the respective transmitting filters.
- (5) FIG. 5 is a sectional view taken along line Z2-Z2 in FIG. 4, illustrating principal components seen in the sectional view.
- (6) FIG. 6 is an explanatory drawing illustrating the transfer of heat generated by the transmitting filters.
- (7) FIG. 7 is a plan view and illustrates the shapes of conductors in Modification 1.
- (8) FIG. 8 is a plan view and illustrates the shapes of conductors in Modification 2.
- (9) FIG. 9 is a plan view and illustrates the shapes of conductors in Modification 3.
- (10) FIG. 10 is a plan view and illustrates the shapes of conductors in Modification 4.
- (11) FIG. 11A is a plan view and illustrates the structures of conductors in Modification 5, and FIG. 11B is a sectional view taken along line Z3-Z3 in FIG. 11A and illustrates principal components seen in the sectional view.
- (12) FIG. 12 is a sectional view and illustrates the structure of conductors in Modification 6.
- (13) FIG. 13A is a plan view and illustrates the layout of conductors in Modification 7, signal lines, and transmitting filters, and FIG. 13B is a sectional view taken along line Z4-Z4 in FIG. 13A.

DETAILED DESCRIPTION

- (14) FIGS. 1 to 13, which will be referred to in relation to, for example, the following embodiment, are schematic views, in which constituent components are not drawn to scale and the size and thickness ratios thereof do not necessarily fully correspond to the actual dimension ratios.

Embodiment

- (15) (1) Overview

(16) Referring to FIGS. 4 and 5, a radio-frequency module 1 according to an embodiment includes a mounting substrate 30 (substrate), a transmitting filter 7A, a transmitting filter 7D, a resin layer 32A, a shield layer 33, a conductor 401, and a conductor 404. The mounting substrate 30 has a first main surface 35 and a second main surface 36 on opposite sides. The transmitting filter 7A is mounted on the first main surface 35 of the mounting substrate 30. The passband of the transmitting filter 7A is a first frequency band. The transmitting filter 7D is mounted on the first main surface 35 of the mounting substrate 30. The passband of the transmitting filter 7D is a second frequency band. The resin layer 32A is disposed on the first main surface 35 of the mounting substrate 30. The transmitting filters 7A and 7D are at least partially covered with the resin layer 32A. The shield layer 33 is disposed on an outer surface of the resin layer 32A. The conductor 401 is disposed on the first main surface 35 of the mounting substrate 30 and is in contact with the transmitting filter 7A and the mounting substrate 30. The conductor 404 is disposed on the first main surface 35 of the mounting substrate 30 and is in contact with the transmitting filter 7B and the mounting substrate 30. The radio-frequency module 1 is capable of performing simultaneous transmission by using both the transmitting filter 7A and the transmitting filter 7D. The conductor 401 extends off an edge of the transmitting filter 7A (a first filter) and to a side other than a side closer to the transmitting filter 7D (a second filter) and is in contact with the shield layer 33. The conductor 404 extends off an edge of the transmitting filter 7A (the first filter) and to a side other than a side closer to the transmitting filter 7A (the second filter) and is in contact with the shield layer 33. The conductors 401 and 404 disposed as above reduce the possibility that

heat generated by the transmitting filters 7A and 7D will cause deterioration of the characteristics of the transmitting filters 7A and 7D while both the transmitting filters 7A and 7D are used for simultaneous transmission.

(17) (2) Details

(18) The radio-frequency module **1** and a communication device **100** according to an embodiment will be described in detail with reference to FIGS. **1** to **6**.

(19) (2-1) Configuration of Communication Device

(20) Referring to FIG. **1**, the communication device **100** includes the radio-frequency module **1**. The communication device **100** is, for example, a mobile terminal, such as a smartphone. In some embodiments, the communication device **100** is a wearable terminal, such as a smart watch. The radio-frequency module **1** supports, for example, the fourth-generation mobile communication (4G) standard or the fifth-generation mobile communication (5G) standard. The 4G standard is, for example, the 3GPP Long Term Evolution (LTE) standard. The 5G standard is, for example, 5G New Radio (NR). The radio-frequency module **1** supports, for example, carrier aggregation and dual connectivity.

(21) The communication device **100** includes, in addition to the radio-frequency module **1**, a signal processing circuit **2**, and a plurality of antennas. For example, three antennas denoted respectively by **3A** to **3C** are included as illustrated in FIG. **1**.

(22) The radio-frequency module **1** is configured to amplify transmission signals (radio-frequency signals) input from the signal processing circuit **2** and to output resultant signals to the antennas **3A** to **3C**. The radio-frequency module **1** is also configured to amplify reception signals input through the antennas **3A** to **3C** and to output resultant signals to the signal processing circuit **2**. The radio-frequency module **1** may be controlled by the signal processing circuit **2**.

(23) The signal processing circuit **2** is connected to the radio-frequency module **1** and is configured to process transmission signals for transmission to the radio-frequency module **1** and to process reception signals received from the radio-frequency module **1**. The transmission signals are to be transmitted through the antennas **3A** to **3C**, and the reception signals are received through the antennas **3A** to **3C**. The signal processing circuit **2** includes a baseband signal processing circuit **21** and a radio-frequency (RF) signal processing circuit **22**.

(24) The baseband signal processing circuit **21** is, for example, a baseband integrated circuit (BBIC). The baseband signal processing circuit **21** generates transmission signals from baseband signals (e.g., audio signals and image signals) input from the outside and outputs resultant transmission signals to the RF signal processing circuit **22**. The baseband signal processing circuit **21** receives reception signals from the RF signal processing circuit **22** and outputs the reception signals to the outside. The signals (reception signals) output by the baseband signal processing circuit **21** are used, for example, as image signals for displaying an image or as audio signals for a telephone conversation.

(25) The RF signal processing circuit **22** is, for example, a radio-frequency integrated circuit (RFIC) and processes radio-frequency signals (transmission signals and reception signals). The RF signal processing circuit **22** performs signal processing such as up-conversion on transmission signals output by the baseband signal processing circuit **21** and outputs resultant signals to the radio-frequency module **1**. The RF signal processing circuit **22** also performs signal processing such as down-conversion on reception signals received from the radio-frequency module **1** and outputs resultant signals to the baseband signal processing circuit **21**.

(26) (2-2) Circuit Configuration of Radio-Frequency Module

(27) As illustrated in FIG. **1**, the radio-frequency module **1** transmits radio-frequency signals (reception signals and transmission signals) between the signal processing circuit **2** and each of the antennas **3A** to **3C**.

(28) The radio-frequency module **1** includes a plurality of power amplifiers, a plurality of low-noise amplifiers, a plurality of transmitting filters, and a plurality of receiving filters. For example,

two power amplifiers respectively denoted by **5A** and **5B**, four low-noise amplifiers respectively denoted by **6A** to **6D**, four transmitting filters respectively denoted by **7A** to **7D**, and four receiving filters respectively denoted by **8A** to **8D** are included, as illustrated in FIG. 1. The radio-frequency module **1** also includes a plurality of output matching circuits and a plurality of matching circuits. For example, two output matching circuits respectively denoted by **10A** and **10B**, four matching circuits respectively denoted by **11A** to **11D**, and three matching circuits respectively denoted by **12A** to **12C** are included, as illustrated in FIG. 1. The radio-frequency module **1** also includes a first switch **14**, a second switch **15**, a third switch **16**, a controller **17**, and a plurality of external connection terminals. For example, eight external connection terminals respectively denoted by **18A** to **18J** are included, as illustrated in FIG. 1. The radio-frequency module **1** also includes a plurality of signal paths for transmission of signals and a plurality of signal paths for reception of signals. For example, two signal paths respectively denoted by **T1** to **T4** are included for transmission of signals, and four signal paths respectively denoted by **R1** to **R4** are included for reception of signals, as illustrated in FIG. 1.

(29) Antenna terminals respectively denoted by **18A** to **18C**, signal output terminals respectively denoted by **18D** to **18F**, and signal input terminals respectively denoted by **18G** to **18I**, and more than one ground terminal denoted by **18J** (see FIG. 3) are included as the external connection terminals **18A** to **18J**. The antennas **3A** to **3C** are connected to the antenna terminals **18A** to **18C**, respectively. Reception signals processed by the radio-frequency module **1** are output to the RF signal processing circuit **22** through the signal output terminals **18D** to **18F**. The signal output terminals **18D** to **18F** are connected to input parts of the signal processing circuit **2** with electric conduction paths therebetween. For example, each of the signal output terminals **18D** to **18F** is connected to the corresponding one of three input parts of the signal processing circuit **2** illustrated in FIG. 1. Transmission signals output by the RF signal processing circuit **22** included in the signal processing circuit **2** are input to the signal input terminals **18G** and **18H**. The signal input terminals **18G** and **18H** are connected to output parts of the RF signal processing circuit **22** with electric conduction paths therebetween. For example, each of the signal input terminals **18G** and **18H** is connected to the corresponding one of two output parts of the RF signal processing circuit **22** illustrated in FIG. 1. Control signals output by the signal processing circuit **2** are input to the signal input terminal **18I**. The signal input terminal **18I** is connected to another output part of the RF signal processing circuit **22** with an electric conduction path therebetween. The external connection terminals **18J** are ground terminals and are hereinafter referred to as ground terminals **18J**. The ground terminals **18J** are connected to ground electrodes of the communication device **100** and are placed at the ground potential accordingly.

(30) The signal path **T1** connects the signal input terminal **18G** and a selection terminal **14h** of the first switch **14** to each other. The power amplifier **5A**, the output matching circuit **10A**, and the transmitting filter **7A** are disposed on the signal path **T1** and are arranged in sequence from the side close to the signal input terminal **18G** to the side close to the first switch **14**. A signal (transmission signal) input to the signal input terminal **18G** is processed by the power amplifier **5A**, the output matching circuit **10A**, and the transmitting filter **7A** while flowing through the signal path **T1**. The resultant signal is then output to the first switch **14**.

(31) The signal paths **T2** to **T4** connect the signal input terminal **18H** to selection terminals **14i** to **14k** of the first switch **14**. The signal paths **T2** to **T4** each includes a common path and an individual path. The common path is denoted by **Tc**, and the individual paths are denoted by **T21**, **T31**, and **T41**, respectively. The common path **Tc** is shared by the signal paths **T2** to **T4**. Each of the individual paths **T21**, **T31**, and **T41** is part of the corresponding one of the signal paths **T2** to **T4**. The common path **Tc** connects the signal input terminal **18H** and a common terminal **15a** of the second switch **15** to each other. The individual paths **T21** to **T41** connect selection terminals **15b** to **15d** of the second switch to the selection terminals **14i** to **14k** of the first switch **14**. More specifically, each of the individual paths **T21** to **T41** connects the corresponding one of the

selection terminals **15b** to **15d** and the corresponding one of the selection terminals **14i** to **14k** to each other. The power amplifier **5B** and the output matching circuit **10B** are disposed on the common path **Tc** and are arranged in sequence from the side close to the signal input terminal **18H** to the side close to the first switch **14**. Each of the transmitting filters **7B** to **7D** is disposed on the corresponding one of the individual paths **T21** to **T41**.

(32) The second switch **15** enables selective switching between three signal paths respectively denoted by **T2** to **T4**. More specifically, the path to which the common path **Tc** is to be connected is selected out of the individual paths **T21** to **T41**. In this way, the second switch **15** performs selective switching between the three signal paths respectively denoted by **T2** to **T4**. The second switch **15** selects one out of the signal paths **T2** to **T4** such that the transmitting filter that is to be used for transmission of signals is selected from among the transmitting filters **7B** to **7C**. In the present embodiment, the second switch **15** selects one out of the signal paths **T2** to **T4**. In some embodiments, however, the second switch **15** selects two or more signal paths.

(33) A signal (transmission signal) input to the signal input terminal **18H** is processed by the power amplifier **5B**, the output matching circuit **10B**, and the transmitting filter selected by the second switch **15** (i.e., one of the transmitting filters **7B** to **7D**) while flowing through the signal path selected by the second switch **15** (i.e., one of the signal paths **T2** to **T4**). The resultant signal is then output to the first switch **14**.

(34) The signal paths **R1** to **R4** connect selection terminals **14d** to **14g** of the first switch **14** to selection terminals **16d** to **16g** of the third switch **16**. The receiving filter **8A**, the matching circuit **11A**, and the low-noise amplifier **6A** are disposed on the signal path **R1** and are arranged in sequence from the side close to the first switch **14** to the side close to the third switch **16**. The same goes for each of the signal paths **R2** to **R4** on which the respective receiving filters (the receiving filters **8B** to **8D**, the respective matching circuits (the matching circuits **11B** to **11D**), and the respective low-noise amplifiers (the low-noise amplifiers **6B** to **6D**) are disposed. A signal (reception signal) output through the selection terminal **14d** of the first switch **14** is processed by the receiving filter **8A**, the matching circuit **11A**, and the low-noise amplifier **6A** while flowing through the signal path **R1**. The resultant signal is then output to the third switch **16**. The same goes for each of the signal paths **R2** to **R4** on which the respective receiving filters (the receiving filters **8B** to **8D**, the respective matching circuits (the matching circuits **11B** to **11D**), and the respective low-noise amplifiers (the low-noise amplifiers **6B** to **6D**) process signals output through the selection terminals **14e** to **14g** of the first switch **14**.

(35) The first switch **14** enables selective switching such that the signal paths **T1** to **T4** and the signal paths **R1** to **R4** are each connected to one of the three antennas respectively denoted by **3A** to **3C**. The first switch **14** is, for example, a switching integrated circuit (IC).

(36) The first switch **14** includes a plurality of common terminals and a plurality of selection terminals. For example, three common terminals respectively denoted by **14a** to **14c** and eight selection terminals respectively denoted by **14d** to **14k** are included as illustrated in FIG. 1. Each of the three common terminals respectively denoted by **14a** to **14c** is connected to the corresponding one of the antenna terminals **18A** to **18C** with the corresponding one of the matching circuits **12A** to **12C** therebetween. That is, each of the three common terminals respectively denoted by **14a** to **14c** is connected the corresponding one of the antennas **3A** to **3C** with the corresponding one of the matching circuits **12A** to **12C** and the corresponding one of the antenna terminals **18A** to **18C** therebetween. Four of the eight selection terminals respectively denoted by **14d** to **14k**, namely, the selection terminals **14d** to **14g** are connected to the four signal paths respectively denoted by **R1** to **R4**. More specifically, each of the selection terminals **14d** to **14g** is connected to an input part of the corresponding one of the receiving filters **8A** to **8D** with the corresponding one of the signal paths **R1** to **R4** therebetween. The other four selection terminals respectively denoted by **14h** to **14k** are connected to the four signal paths respectively denoted by **T1** to **T4**. More specifically, each of the selection terminals **14h** to **14k** is connected to an output part of the corresponding one of the

transmitting filters 7A to 7D with the corresponding one of the signal paths T1 to T4 therebetween.

(37) The first switch **14** makes a selection in accordance with a control signal output by the controller **17** to connect each common terminal to a desired selection terminal (signal path); that is, for each of the three common terminals respectively denoted by **14a** to **14c** (i.e., the three antennas respectively denoted by 3A to 3C), the first switch **14** makes a selection from the eight selection terminals respectively denoted by **14d** to **14k** (i.e., the eight signal paths respectively denoted by T1 to T4 and R1 to R4). For example, the first switch **14** is capable of establishing one-to-one connections and one-to-many connections.

(38) The second switch **15** is a switch for selecting, out of the signal paths T2 to T4, a signal path through which transmission signals are to be transmitted. The second switch **15** is, for example, a switching IC. The second switch **15** includes the common terminal **15a** and a plurality of selection terminals. For example, three selection terminals respectively denoted by **15b** to **15d** are included as illustrated in FIG. 1. The common terminal **15a** is connected to the common path Tc. More specifically, the common terminal **15a** is connected to an output part of the output matching circuit **10B** with the common path Tc therebetween. Each of the three selection terminals respectively denoted by **15b** to **15d** is connected to the corresponding one of the three signal paths respectively denoted by T2 to T4. More specifically, each of the selection terminals **15b** to **15d** is connected to an input part of the corresponding one of the transmitting filters 7B to 7D with the corresponding one of the signal paths T2 to T4 therebetween.

(39) The second switch **15** makes a selection in accordance with a control signal output by the controller **17** to connect the common terminal **15a** to a desired selection terminal; that is, the second switch **15** makes a selection from the three selection terminals respectively denoted by **15b** to **15d**. In this way, the second switch **15** selects, out of the signal paths T2 to T4, a signal path through which transmission signals are to be transmitted. In the present embodiment, the second switch **15** establishes a one-to-one connection. In some embodiments, however, the second switch **15** establishes a one-to-many connection.

(40) The third switch **16** is a switch for selecting, out of the signal paths R1 to R4, a signal path through which reception signals are to be transmitted. The third switch **16** is, for example, a switching IC. The third switch **16** includes a plurality of common terminals and a plurality of selection terminals. For example, three common terminals respectively denoted by **16a** to **16c** and four selection terminals respectively denoted by **16d** to **16g** are included as illustrated in FIG. 1. Each of the three selection terminals respectively denoted by **16a** to **16c** is connected to the corresponding one of the three signal output terminals respectively denoted by **18D** to **18F**. Each of the four selection terminals respectively denoted by **16d** to **16g** is connected to the corresponding one of the four signal paths respectively denoted by R1 to R4. More specifically, each of the selection terminals **16d** to **16f** is connected to an output part of the corresponding one of the low-noise amplifiers 6A to 6D with the corresponding one of the signal paths R1 to R4 therebetween.

(41) The third switch **16** makes a selection in accordance with a control signal output by the controller **17** to connect each common terminal to a desired selection terminal (signal path); that is, for each of the three common terminals respectively denoted by **16a** to **16c**, the third switch **16** makes a selection from the four selection terminals respectively denoted by **16d** to **16f** (i.e., the four signal paths respectively denoted by R1 to R4). When being connected to one of the common terminals **16a** to **16c**, each of the signal paths R1 to R4 is selected as a signal path that is to be used. The third switch **16** may establish one-to-one connections or one-to-many connections.

(42) The power amplifiers 5A and 5B are disposed on the signal paths T1 and T2, respectively. More specifically, the power amplifier 5B is disposed on the common path Tc that is part of the signal path T2. The power amplifiers 5A and 5B each has an input part and an output part. The input part of the power amplifier 5A and the input part of the power amplifier 5B are connected to the signal input terminals **18G** and **18H**, respectively. The output part of the power amplifier 5A and the output part of the power amplifier 5B are connected to an input part of the output matching

circuit **10A** and an input part of the output matching circuit **10B**. The power amplifiers **5A** and **5B** amplify transmission signals input to the respective input parts and then output resultant signals from the respective output parts. The power amplifiers **5A** and **5B** are controlled by the controller **17**.

(43) The low-noise amplifiers **6A** to **6D** are disposed on the signal paths **R1** to **R4**, respectively. The low-noise amplifiers **6A** to **6D** each has an input part and an output part. The input parts of the low-noise amplifiers **6A** to **6D** are connected to output parts of the matching circuits **11A** to **11D**, respectively. The output parts of the low-noise amplifiers **6A** to **6D** are connected to the selection terminals **16d** to **16g** of the third switch **16**. The low-noise amplifiers **6A** to **6D** amplify reception signals input to the respective input parts and then output resultant signals from the respective output parts. The low-noise amplifiers **6A** to **6D** are controlled by controller **17**.

(44) The transmitting filters **7A** and **7D** are disposed on the signal paths **T1** to **T4**, respectively. More specifically, the transmitting filters **7B** to **7D** are disposed on the individual path **T21** being part of the signal path **T2**, the individual path **T31** being part of the signal path **T3**, and the individual path **T41** being part of the signal path **T4**, respectively. The transmitting filters **7A** to **7D** are filters whose passbands are different communication bands. For example, the passbands of the transmitting filters **7A** to **7D** are the transmission band of a first communication band (first frequency band), the transmission band of a second communication band, the transmission band of a third communication band, and the transmission band of a fourth communication band (second frequency band), respectively. The transmitting filter **7A** to **7D** each has an input part and an output part. The input part of the transmitting filter **7A** is connected to an output part of the output matching circuit **10A**. The output part of the transmitting filter **7A** is connected to the selection terminal **14h** of the first switch **14**. The input parts of the transmitting filters **7B** to **7D** are connected to the selection terminals **15b** to **15d** of the second switch **15**, respectively. The output parts of the transmitting filters **7B** to **7D** are connected to the selection terminals **14i** to **14k** of the first switch **14**, respectively. A transmission signal input to the input part of each of the transmitting filters **7A** to **7D** is band-limited to the transmission band of the corresponding one of the first to fourth communication bands and is then output from the output part.

(45) The transmitting filter **7A** is connected to the power amplifier **5A** with the output matching circuit **10A** therebetween. Each of the transmitting filters **7B** to **7D** is connected to the power amplifier **5B** with the second switch **15** and the output matching circuit **10B** therebetween. That is, the transmitting filter **7A** and each of the transmitting filters **7B** to **7D** are connected to different power amplifiers, which are denoted by **5A** and **5B**, respectively.

(46) The receiving filters **8A** to **8D** are disposed on the signal paths **R1** to **R4**, respectively. The receiving filters **8A** to **8D** are filters whose passbands are different communication bands. For example, the passbands of the receiving filters **8A** to **8D** are the reception band (communication band) of the first communication band, the reception band (communication band) of the second communication band, the reception band (communication band) of the third communication band, and the reception band (communication band) of the fourth communication band, respectively. The receiving filters **8A** to **8D** each has an input part and an output part. The input parts of the receiving filters **8A** to **8D** are connected to the selection terminals **14d** to **14f** of the first switch **14**, respectively. The output parts of the receiving filters **8A** to **8D** are connected to input parts of the matching circuits **11A** to **11D**, respectively. A reception signal input to the input part of each of the receiving filters **8A** to **8D** is band-limited to the reception band of the corresponding one of the first to fourth communication bands and is then output from the output part.

(47) Each of the receiving filters **8A** to **8D** is connected to the corresponding one of the low-noise amplifiers **6A** to **6D** with the corresponding one of the matching circuits **11A** to **11D** therebetween. That is, the receiving filters **8A** to **8D** are connected to different low-noise amplifiers, which are denoted by **6A** to **6D**, respectively.

(48) The transmitting filters **7A** to **7D** and the receiving filters **8A** to **8D** are, for example, acoustic

wave filters. For example, the acoustic wave filters are surface acoustic wave (SAW) filters that use surface acoustic waves. It is not required that the transmitting filters **7A** to **7D** and the receiving filters **8A** to **8D** be SAW filters. In some embodiments, the transmitting filters **7A** to **7D** and the receiving filters **8A** to **8D** are bulk acoustic wave (BAW) filters.

(49) The output matching circuit **10A** is disposed on the signal path **T1**. The output matching circuit **10B** is disposed on the common path **Tc**. The output matching circuit **10A** is a circuit that provides impedance matching between the power amplifier **5A** and the transmitting filter **7A** and includes, for example, an inductor. The output matching circuit **10A** is connected between the output part of the power amplifier **5A** and the input part of the transmitting filter **7A**. The output matching circuit **10B** is a circuit that provides impedance matching between the power amplifier **5B** and each of the transmitting filters **7B** to **7D** and includes, for example, an inductor. The output matching circuit **10B** is connected between the output part of the power amplifier **5A** and the common terminal **15a** of the second switch **15**.

(50) The matching circuits **11A** to **11D** are disposed on the signal paths **R1** to **R4**, respectively. Each of the matching circuits **11A** to **11D** is a circuit that provides impedance matching between the corresponding one of the low-noise amplifiers **6A** to **6D** and the corresponding one of the receiving filters **8A** to **8D** and includes, for example, an inductor. Each of the matching circuits **11A** to **11D** is connected between the output part of the corresponding one of the receiving filters **8A** to **8D** and the input part of the corresponding one of the low-noise amplifiers **6A** to **6D**. Each of the matching circuits **12A** to **12C** is connected between the corresponding one of the common terminals **14a** to **14c** of the first switch **14** and the corresponding one of the antenna terminals **18A** to **18C**. Each of the matching circuits **12A** to **12C** is a circuit that provides impedance matching between the first switch **14** and the corresponding one of the antennas **3A** to **3C** and includes, for example, an inductor.

(51) The controller **17** is a control device that controls circuit components (e.g., the power amplifiers **5A** and **5B**, the low-noise amplifiers **6A** to **6D**, the first switch **14**, the second switch **15**, and the third switch **16**) in accordance with control signals output by the signal processing circuit **2**. The controller **17** is electrically connected to the circuit components, such as the power amplifiers **5A** and **5B** and the low-noise amplifiers **6A** to **6D**. The controller **17** is connected the output part of the signal processing circuit **2** with the signal input terminal **18I** therebetween. The controller **17** controls the circuit components in accordance with control signals output by the signal processing circuit **2** and input to the signal input terminal **18I**.

(52) (2-3) Workings of Communication Device

(53) The workings of the communication device **100** will be described below with reference to FIG. **1**.

(54) The communication device **100** can transmit transmission signals in two communication bands (e.g., the first communication band and one of the second to fourth communication bands) simultaneously. To that end, the second switch **15** selects, out of the signal paths **T2** to **T4**, a signal path through which transmission signals are to be transmitted. For example, the signal path **T2** is selected. The first switch **14** selects, out of the three antennas respectively denoted by **3A** to **3C**, an antenna that is to be connected to the signal path **T1** and an antenna that is to be connected to the signal path selected by the second switch **15**. For example, the antenna **3B** is selected for connection to the signal path **T1**, and the antenna **3C** is selected for connection to the signal path **T2**.

(55) In this state, transmission signals are output from the two of the output parts of the signal processing circuit **2** to two of the signal input terminals of the radio-frequency module **1** or, more specifically, to the signal input terminals **18G** and **18H**. A transmission signal input to the signal input terminal **18G** is processed by the processing units (the power amplifier **5A**, the output matching circuit **10A**, and the transmitting filter **7A**) while flowing through the signal path **T1**. The resultant signal flows through the first switch **14** and the matching circuit **12B** and is then

transmitted to the outside through the antenna **3B**. A transmission signal input to the signal input terminal **18H** is processed by the processing units (the power amplifier **5B**, the output matching circuit **10B**, and the transmitting filter **7B**) while flowing through the signal path **T2** selected by the second switch **15**. The resultant signal flows through the first switch **14** and the matching circuit **12C** and is then transmitted to the outside through the antenna **3C**.

(56) The communication device **100** transmits transmission signals in two communication bands (e.g., the first communication band and one of the second to fourth communication bands) simultaneously, in which case two transmitting filters corresponding to the two communication bands (e.g., the transmitting filter **7A** and one of the transmitting filters **7B** to **7D**) work concurrently. That is, the two transmitting filters are used to enable simultaneous transmission.

(57) Although simultaneous transmission of transmission signals in two transmission bands is herein described as an embodiment, the present disclosure is applicable to simultaneous transmission of transmission signals in three or more transmission bands. In some embodiments, three or more transmitting filters are used to enable simultaneous transmission of transmission signals in three or more corresponding transmission bands.

(58) The communication device **100** can receive reception signals in two transmission bands (e.g., two of the first to fourth communication bands) simultaneously. To that end, the third switch **16** selects, out of the signal paths **R1** to **R4**, two signal paths through which reception signals are to be transmitted. For example, the signal paths **R1** and **R2** are selected. For example, the common terminals **16a** and **16b** of the third switch **16** connect the selected two signal paths, namely, the signal paths **R1** and **R2** to the signal output terminals **18D** and **18E**, respectively. The first switch **14** selects, out of the three antennas respectively denoted by **3A** to **3C**, an antenna that is to be connected to the signal path selected by the third switch **16**. For example, the antenna **3A** is selected for connection to the signal paths **R1** and **R2**.

(59) A reception signal received through the antenna **3A** flows through the matching circuit **12A** and the first switch **14**. The first switch **14** splits the signal into two. In a state in which the two signal paths respectively denoted by **R1** and **R2** are selected by the third switch **16**, the resultant reception signals flow through the signal paths **R1** and **R2**, respectively. The reception signal flowing through the signal path **R1** is output to one of the input parts of the signal processing circuit **2** via the receiving filter **8A**, the matching circuit **11A**, the low-noise amplifier **6A**, the third switch **16**, and the signal output terminal **18D**. The reception signal flowing through the signal path **R2** is output to one of the input parts of the signal processing circuit **2** via the receiving filter **8B**, the matching circuit **11B**, the low-noise amplifier **6B**, the third switch **16**, and the signal output terminal **18E**. The reception signals output to the input part of the signal processing circuit **2** are processed by the signal processing circuit **2** and are then output to the outside.

(60) The communication device **100** receives reception signals in two communication bands (e.g., the first communication band and the second communication band) simultaneously, in which case two receiving filters corresponding to the two communication bands (e.g., the receiving filter **8A** and the receiving filter **8B**) work concurrently. That is, the two receiving filters are used to enable simultaneous transmission.

(61) Although simultaneous reception of reception signals in two communication bands is herein described as an embodiment, the present disclosure is applicable to simultaneous reception of reception signals in three or more communication bands.

(62) (2-4) Structure of Radio-Frequency Module

(63) The following describes the structure of the radio-frequency module **1** with reference to FIGS. **2** and **3**.

(64) Referring to FIGS. **2** and **3**, the radio-frequency module **1** includes the mounting substrate **30** (substrate), circuit components **31**, the resin layer **32A**, a resin layer **32B**, and the shield layer **33**.

(65) The mounting substrate **30** is a substrate that is to be populated with circuit components as illustrated in FIG. **3**. For example, the mounting substrate **30** is in the form of a rectangular plate.

The mounting substrate **30** has the first main surface **35** and the second main surface **36**, which are located on opposite sides in the thickness direction of the mounting substrate **30**. The thickness direction is denoted by **D1**. The first main surface **35** and the second main surface **36** are, for example, rectangular (see FIG. 2).

(66) The mounting substrate **30** is a multilayer substrate including, for example, a plurality of dielectric layers and a plurality of electrically conductive layers. The dielectric layers and the electrically conductive layers are stacked in the thickness direction **D1** of the mounting substrate **30**. The electrically conductive layers are formed into the respective predetermined patterns. One of the electrically conductive layers is a ground layer. The mounting substrate **30** of the radio-frequency module **1** includes via conductors, each of which forms an electrical connection between the ground layer and the corresponding one of the ground terminals **18J**. The mounting substrate **30** is, for example, a low-temperature co-fired ceramic (LTCC) substrate. It is not required that the mounting substrate **30** be an LTCC substrate. The mounting substrate **30** may be a printed wiring board, a high-temperature co-fired ceramic (HTCC) substrate, or a resin multilayer substrate.

(67) The thickness direction **D1** of the mounting substrate **30** is hereinafter also referred to as a first direction **D1**. **D2** denotes a second direction that is orthogonal to the first direction **D1**. For example, the second direction **D2** is parallel to one of two pairs of opposite edges of the first main surface **35** of the mounting substrate **30**. **D3** denotes a third direction that is orthogonal to both the first direction **D1** and the second direction **D2**. For example, the second direction **D2** is parallel to the other pair of opposite edges of the first main surface **35** of the mounting substrate **30**.

(68) The upper and lower sides that are opposite sides in the second direction **D2** on the drawing plane of FIG. 2 may be simply referred to as the rear side and the front side, respectively. Similarly, the left and right sides that are opposite sides in the third direction **D3** on the drawing plane of FIG. 2 may be simply referred to as the left side and the right side, respectively. The upper and lower sides that are opposite sides in the first direction **D1** on the drawing plane of FIG. 3 may be simply referred to as the upper side and the lower side, respectively.

(69) Some of the circuit components **31** are mounted on the first main surface **35** of the mounting substrate **30**, and the other circuit components **31** are mounted on the second main surface **36** of the mounting substrate **30**. This implies, for example, that the circuit components **31** are each disposed on (i.e., mechanically connected to) the first main surface **35** or the second main surface **36** of the mounting substrate **30** or are electrically connected to (the relevant conductor portions of) the mounting substrate **30**.

(70) Examples of the circuit components **31** include the controller **17**, the power amplifiers **5A** and **5B**, the transmitting filters **7A** to **7D**, the receiving filters **8A** to **8D**, the output matching circuits **10A** and **10B**, the matching circuits **11A** to **11D**, and the matching circuits **12A** to **12C** (see FIG. 2). The circuit components mentioned above are mounted on the first main surface **35** of the mounting substrate **30**. The outside shape of each of the circuit components viewed in plan (i.e., in the thickness direction **D1** of the mounting substrate **30**) is, for example, rectangular.

(71) The first main surface **35** of the mounting substrate **30** includes a first half region **35L** and a second half region **35R**. The first half region **35L** of the first main surface **35** is located on one side or, more specifically, the left side (i.e., a first side) in the third direction **D3**. The second half region **35R** of the first main surface **35** is located on the other side, namely, the right side (i.e., a second side) in the third direction **D3**.

(72) Circuit components disposed on the first half region **35L** of the mounting substrate **30** are mainly those relevant to signal transmission; that is, the power amplifiers **5A** and **5B**, the transmitting filters **7A** to **7D**, the output matching circuits **10A** and **10B**, and the controller **17** are disposed on the first half region **35L**. The controller **17** and the transmitting filter **7A** are disposed on one side of the first half region **35L** or, more specifically, on one side (the upper side) in the second direction **D2**. The controller **17** and the transmitting filter **7A** are located one side (the left side) and the other side (the right side), respectively, in the third direction **D3**. The power amplifiers

5A 5B and the output matching circuit 10A are located in the midsection of the first half region 35L in the second direction D2 and are arranged in sequence in the third direction D3. The power amplifier 5A and the output matching circuit 10A are located on one side (the left side) and the other side (the right side), respectively, in the third direction D3. The output matching circuit 10B, the transmitting filter 7B, the transmitting filter 7C, and the transmitting filter 7D are disposed on another side of the first half region 35L, namely, the other side (the lower side) in the second direction D2 and are arranged in sequence in the third direction D3. The output matching circuit 10B and the transmitting filter 7D are on one side (the left side) and the other side (the right side), respectively, in the third direction D3.

(73) Components disposed on the second half region 35R of the mounting substrate 30 are mainly those relevant to signal reception; that is, the receiving filters 8A to 8D, the matching circuits 11A to 11D, and the matching circuits 12A to 12C are disposed on the second half region 35R. The receiving filters 8A to 8D are located in the midsection of the second half region 35R and are arranged in a matrix with two rows and two columns. The matching circuit 11A and the matching circuit 11B are disposed on one side of the second half region 35R or, more specifically, on the upper side in the second direction D2 and are arranged side by side in the third direction D3. The matching circuit 11C and the matching circuit 11D are disposed on another side of the second half region 35R or, more specifically, on the right side in the third direction D3 and are arranged side by side in the second direction D2. The matching circuits 12A to 12C are disposed on another side of the second half region 35R or, more specifically, on the lower side in the second direction D2 and are arranged side by side in the third direction D3.

(74) As described above, the circuit components relevant to signal transmission are disposed on the first half region 35L, and the circuit components relevant to signal reception are disposed on the second half region 35R. The division of the mounting region eliminates or reduces the possibility of interference between the circuit components relevant to signal transmission and the circuit components relevant to signal reception, that is, interference between transmission signals and reception signals.

(75) Examples of the circuit components 31 also include the first switch 14, the second switch 15, the third switch 16, and the low-noise amplifier 6A to 6D. The circuit components 31 mentioned above are mounted on the second main surface 36 of the mounting substrate 30.

(76) As illustrated in FIG. 3, the resin layer 32A (hereinafter also referred to as “first resin layer 32A”) is disposed on the first main surface 35 of the mounting substrate 30. The circuit components 31 disposed on the first main surface 35 of the mounting substrate 30 are covered with the first resin layer 32A. More specifically, the transmitting filters 7A to 7B, which are capable of working concurrently for simultaneous transmission of signals, are at least partially covered with the resin layer 32A. The circuit components 31 mentioned above are sealed with the first resin layer 32A. The first resin layer 32A is not illustrated in FIG. 2.

(77) The resin layer 32B (hereinafter also referred to as “second resin layer 32B”) is disposed on the second main surface 36 of the mounting substrate 30. The circuit components 31 mounted on the second main surface 36 of the mounting substrate 30 are covered with the second resin layer 32B and are sealed with the second resin layer 32B. More specifically, the external connection terminals 18A to 18J on the second main surface 36 are covered with the second resin layer 32B, with their tip faces being exposed at the surface of the second resin layer 32B. The circuit components 31 mounted on the second main surface 36 are entirely covered with the second resin layer 32B. The second resin layer 32B contains resin. The second resin layer 32B may be made of a mixture of resin and filler. The second resin layer 32B and the first resin layer 32A may be made of the same material or made be made of different materials.

(78) The shield layer 33 is made of, for example, metal. The shield layer 33 is disposed on the outer surface (a peripheral surface and a top surface) of the first resin layer 32A, a peripheral surface of the second resin layer 32B, and a peripheral surface of the mounting substrate 30. The outer surface

of the first resin layer **32A** and the peripheral surface of the mounting substrate **30** are entirely covered with the shield layer **33**. The peripheral surface of the second resin layer **32B** is partially covered with the shield layer **33**. More specifically, one part of the peripheral surface closer to the mounting substrate **30** than the other part in the first direction **D1** is covered with the shield layer **33**. The shield layer **33** is in contact with the ground layer of the mounting substrate **30**. The shield layer **33** and the ground layer are placed at the same potential (at the ground potential of the radio-frequency module **1**) accordingly.

(79) (2-5) Heat Transfer Paths for Transmitting Filters

(80) As described above with regard to the radio-frequency module **1** (see FIG. 2), the circuit components **31** relevant to signal transmission are disposed on the first half region **35L**, and the circuit components **31** relevant to signal reception are disposed on the second half region **35R**. In the present embodiment, any two or more of the transmitting filters **7A** to **7D** can work concurrently for simultaneous transmission of signals. For example, the transmitting filters **7A** and **7D** work concurrently for simultaneous transmission of signals. That is, the transmitting filters **7A** and **7D** are used to enable simultaneous transmission. The transmitting filters **7A** and **7D** that are capable of working concurrently for simultaneous transmission of signals are gathered together in the first half region **35L** and are therefore close to each other. In the present embodiment, well thought-out heat transfer paths are provided to eliminate or reduce the possibility that the transmitting filters **7A** and **7D** that are close to each other and capable of working concurrently will affect each other due to heat generation. The well through-out heat transfer paths provided respectively for the transmitting filter **7A** and **7D** will be described in detail below with reference to FIGS. 4 and 5.

(81) Referring to FIG. 4, the radio-frequency module **1** includes the conductors **401** and **404**, which are disposed on the first main surface **35** of the mounting substrate **30** and are provided for the transmitting filters **7A** and **7D**, respectively. Each of the conductors **401** and **404** is a heat transfer path for the corresponding one of the transmitting filters **7A** and **7D**. Only some of the circuit components **31** disposed on the first main surface **35** or, more specifically, only the transmitting filters **7A** and **7D** and the conductors **401** and **404** are illustrated in FIG. 4.

(82) As illustrated in FIG. 4, the transmitting filters **7A** and **7D** on the first main surface **35** of the mounting substrate **30** are located in the midsection in the third direction **D3** and are close to the respective edges that are opposite in the second direction **D2**. The conductor **401** serving as a heat transfer path for the transmitting filter **7A** and the conductor **404** serving as a heat transfer path for the transmitting filter **7D** are disposed on the first main surface **35** of the mounting substrate **30**. The wiring conductors of the mounting substrate **30** include wiring conductors other than the conductors **401** and **404**. In some embodiments, however, the wiring conductors of the mounting substrate **30** serve as the conductors **401** and **404**.

(83) Each of the conductors **401** and **404** is in contact with the mounting substrate **30** and the corresponding one of the transmitting filters **7A** and **7D** (see FIG. 5). Each of the conductors **401** and **404** extends off an edge of the corresponding one of the transmitting filters **7A** and **7D** and to a side other than a side closer to the other transmitting filter (the transmitting filter **7D** or **7A**) and is in contact with the shield layer **33** (see FIG. 4). That is, each of the conductor **401** and **404** provided respectively for the transmitting filters **7A** and **7D** is in contact with the shield layer **33** on a side other than the side closer to the other transmitting filter (the transmitting filter **7D** or **7A**) than to the transmitting filter concerned. More specifically, the conductors **401** and **402** are provided for the transmitting filters **7A** and **7D**, respectively. The conductor **401** provided for the transmitting filter **7A** is in contact with the shield layer **33** on a side other than the side closer to the transmitting filter **7D** than to the transmitting filter **7A**. The conductor **404** provided for the transmitting filter **7B** is in contact with the shield layer **33** on a side other than the side closer to the transmitting filter **7A** than to the transmitting filter **7B**.

(84) The shield layer **33** in an example illustrated in FIGS. 4 and 5 has surface portions with which

the outer surface (the top surface and four side surfaces) of the resin layer 32A is covered. More specifically, the shield layer 33 has a top surface denoted by 331 and four side surfaces respectively denoted by 332 to 335. The five surfaces of the shield layer 33, that is, the top surface 331 and the four side surfaces respectively denoted by 332 to 335 may be collectively referred to as surface portions when there is no need to distinguish one from another; that is, the five surfaces may be hereinafter also referred to as surface portions 331 to 335, respectively.

(85) More specifically, when viewed in plan in the thickness direction of the mounting substrate 30 (i.e., in the first direction D1) as illustrated in FIG. 4, each of the conductors 401 and 404 extends from the corresponding one of the transmitting filters 7A and 7D to the side opposite to the side on which the other transmitting filter (the transmitting filter 7D or 7A) is disposed. The conductors 401 and 404 are in contact with two of the four side surfaces of the shield layer 33 that are respectively denoted by 332 to 335. More specifically, the conductors 401 and 404 are in contact with the side surfaces located on the respective sides to which the conductors 401 and 404 extend. For example, the conductors 401 and 404 are in contact with the side surfaces 332 and 333, respectively. The conductor 401 provided for the transmitting filter 7A extends upward from the transmitting filter 7A and is in contact with the side surface 332 located on the upper side of the shield layer 33. The conductor 404 provided for the transmitting filter 7B extends downward from the transmitting filter 7B and is in contact with the side surface 333, which is located on the lower side of the shield layer 33.

(86) That is, with the shield layer 33 having four side surfaces respectively denoted by 332 to 335, the conductors 401 and 404 illustrated in FIG. 4 extend to and are in contact with the side surfaces 332 and 333, which are closer than the other side surfaces of the shield layer 33 to the transmitting filters 7A to 7D. With the shield layer 33 having four side surfaces respectively denoted by 332 to 335, each of the transmitting filters 7A and 7D in the present embodiment is closer to the corresponding one of the side surfaces 332 and 333 than to the other side surfaces of the shield layer 33. The side surfaces 332 and 333 are located on the rear side and the front side, respectively. With the shield layer 33 having four side surfaces respectively denoted by 332 to 335, the conductors 401 and 404 extend to and are in contact with the side surfaces 332 and 333, which are closer than the other side surfaces of the shield layer 33 to the transmitting filters 7A to 7D.

(87) As mentioned above, the shield layer 33 has four side surfaces respectively denoted by 332 to 335, and two of the four side surfaces are closer than the other side surfaces of the shield layer 33 to the transmitting filters 7A and 7D. More specifically, a perpendicular line drawn from the center of the transmitting filter 7A to the corresponding one of the two side surface and a perpendicular line drawn from the center of the transmitting filter 7D to the corresponding one of the two side surfaces are shorter than perpendicular lines drawn from the center of the transmitting filter 7A or 7D to the other side surfaces when viewed in plan in the thickness direction of the mounting substrate 30 (i.e., in the first direction D1). As mentioned above, the conductors 401 and 404 extend to the side surfaces 332 and 333, which are closer than the other side surfaces of the shield layer 33 to the transmitting filters 7A to 7D. More specifically, the conductor 401 extends along the perpendicular line drawn from the center the transmitting filter 7A to the side surface 332 that is closer than the other side surfaces to the transmitting filter 7A, and the conductor 404 extends along the perpendicular line drawn from the center of the transmitting filter 7D to the side surface 333 that is closer than the other side surfaces to the transmitting filter 7D. When viewed in plan in the thickness direction of the mounting substrate 30 (i.e., in the first direction D1), the conductors 401 and 404 in contact with the respective closest side surfaces (i.e., the side surfaces 332 and 333) extend perpendicularly to the respective closest side surfaces (i.e., the side surfaces 332 and 333).

(88) More specifically, when viewed in section in the thickness direction of the mounting substrate 30 (i.e., in the first direction D1) as illustrated in FIG. 5, the conductors 401 and 404 are in contact with the first main surface 35 of the mounting substrate 30 and extend along the first main surface 35. Each of the conductors 401 and 404 on the first main surface 35 of the mounting substrate 30

extending as above also lies in the region in which the corresponding one of the transmitting filters 7A and 7D is placed (i.e., in the region onto which the shape of the corresponding one of the transmitting filters 7A and 7D is projected). Each of the conductors 401 and 404 is in contact with a back surface of the corresponding one of the transmitting filters 7A and 7D with a connection member 41 therebetween. For example, the connection member 41 is a solder bump. More specifically, the transmitting filters 7A and 7D each includes a support substrate 71 and an IDT electrode 71c, which is supported by the support substrate 71. The support substrates 71 of the transmitting filters 7A and 7D mounted on the mounting substrate 30 are each located between the mounting substrate 30 and the IDT electrode 71c. Each of the conductors 401 and 404 is in contact with the support substrate 71 of the corresponding one of the transmitting filters 7A and 7D with the connection member 41 therebetween.

(89) The configuration of the transmitting filters 7A and 7D and the mounting of the transmitting filters 7A and 7D on the mounting substrate 30 will be elaborated with reference to FIG. 5. FIG. 5 is a sectional view of the transmitting filter 7A. The transmitting filters 7A 7D have the same structure and are mounted identically. Thus, FIG. 5 may also be regarded as a sectional view of the transmitting filter 7D. The same holds for the transmitting filters 7B and 7C; that is, the transmitting filters 7A to 7D have the same structure and are mounted identically. The transmitting filters 7A and 7D in the present embodiment each includes, in addition to the support substrate 71, a spacer layer 72, a cover member 73, a plurality of through-via electrodes 74, and a plurality of connection terminals 75. For example, the connection terminals 75 are solder bumps.

(90) One of two main surfaces of the support substrate 71 is denoted by 71a, and circuitry (a functional unit) including the IDT electrode (comb-shaped electrode) 71c and a plurality of pad electrodes 71d are disposed on the main surface 71a. That is, the support substrate 71 supports the IDT electrode 71c. The pad electrodes 71d is electrically connected to the circuitry to form an electrical connection between the circuitry and an external substrate (e.g., the mounting substrate 30). The spacer layer 72 is a member disposed between the support substrate 71 and the cover member 73 and defining a space in which the circuitry including the IDT electrode 71c is placed. For example, the spacer layer 72 is made of an electrical insulating material, such as an epoxy resin or polyimide, and is in the form of a rectangular frame. The spacer layer 72 on the main surface 71a of the support substrate 71 surrounds the circuitry including the IDT electrode 71c.

(91) The cover member 73 is made of an electrical insulating material, such as an epoxy resin or polyimide, and is flat and plate-like in shape. The cover member 73 is disposed on the spacer layer 72, whose top is partially open and is covered with the cover member 73. Each of the through-via electrodes 74 is a conductive member forming an electrical connection between the corresponding one of the pad electrodes 71d on the main surface 71a of the support substrate 71 and the corresponding one of the connection terminals 75 on a main surface 71b of the support substrate 71 and extends across the support substrate 71 in the thickness direction. The connection terminals 75 are conductive members each forming an electrical connection between the support substrate 71 and a wiring conductor of the mounting substrate 30. For example, the connection terminals 75 are solder bumps. Each of the connection terminals 75 and the corresponding one of the through-via electrodes 74 are in contact with each other on the main surface 71b of the support substrate 71, and each of the connection terminals 75 and the wiring conductor of the mounting substrate 30 are in contact with each other on the first main surface 35 of the mounting substrate 30.

(92) The transmitting filters 7A and 7D are mounted on the mounting substrate 30 with the support substrate 71 being closer than the IDT electrode 71c to the mounting substrate 30. Heat generated by the transmitting filters 7A and 7D arises from the support substrates 71 of the respective transmitting filters. More specifically, the IDT electrode 71c generates heat, and the support substrate 71 generates heat accordingly. This is the reason why the transmitting filters 7A and 7D are mounted on the mounting substrate 30 with the support substrate 71 facing the mounting substrate 30. This placement enables an effective heat transfer from each of the transmitting filters

7A and 7D to the corresponding one of the conductors **401** and **404** through the connection member **41**.

(93) The same holds for the transmitting filters **7B** and **7C**; that is, the transmitting filters **7A** to **7D** have the same structure and are mounted identically on the mounting substrate **30**.

(94) FIG. **6** illustrates a case in which the transmitting filters **7A** and **7D** work concurrently for simultaneous transmission of signals and generate heat at the same time. Heat generated by the transmitting filter **7A** and heat generated by the transmitting filter **7D** are denoted by **Q1** and **Q2**, respectively. The heat **Q1** generated by the transmitting filter **7A** is transferred by the corresponding conductor (i.e., the conductor **401**) and flows from the transmitting filter **7A** to the side other than the side closer to the transmitting filter **7D** than to the transmitting filter **7A**. The heat **Q1** is transferred to the shield layer **33** accordingly and is then dissipated from the shield layer **33**. Likewise, the heat **Q2** generated by the transmitting filter **7D** is transferred by the corresponding conductor (i.e., the conductor **404**) and flows from the transmitting filter **7D** to the side other than the side closer to the transmitting filter **7A** than to the transmitting filter **7D**. The heat **Q2** is transferred to the shield layer **33** accordingly and is then dissipated from the shield layer **33**. Each of the conductors **401** and **404** serves as a heat transfer path for the corresponding one of the transmitting filters **7A** and **7D**. Thus, the heat transfer paths for the transmitting filters **7A** and **7D**, that is, the conductors **401** and **404** are less susceptible to the heat **Q2** generated by the transmitting filter **7D** and the heat **Q1** generated by the transmitting filter **7A**, respectively. This yields an improvement in the heat dissipation properties of each of the transmitting filters **7A** and **7D** and thus eliminates or reduces the possibility that heat generated at the same time by the transmitting filters **7A** and **7D** working concurrently for simultaneous transmission of signals will cause deterioration of the characteristics of the transmitting filters **7A** and **7D**.

(95) The resultant improvement in the heat dissipation properties of the transmitting filters **7A** and **7D** eliminates or reduces the possibility that the heat **Q1** generated by the transmitting filter **7A** and the heat **Q2** generated by the transmitting filter **7D** will affect the transmitting filter **7D** and the transmitting filter **7A**, respectively. Thus, the resultant improvement in the heat dissipation properties of one of the two transmitting filters leads to an improvement in the heat dissipation properties of the other transmitting filter. This synergy eliminates or reduces the possibility that heat generated at the same time by the transmitting filters **7A** and **7D** working concurrently for simultaneous transmission of signals will cause deterioration of the characteristics of the transmitting filters **7A** and **7D**. That is, deterioration of the characteristics of each of the transmitting filters **7A** and **7D** is inhibited.

(96) Two of the transmitting filters **7A** to **7D** or, more specifically, the transmitting filters **7A** and **7D** have been described above as examples of the transmitting filters that can work concurrently for simultaneous transmission of signals. The radio-frequency module **1** in the present embodiment also includes conductors (not illustrated) provided for the transmitting filters **7B** and **7C** as well as the conductors **401** and **404** provided for the transmitting filters **7A** and **7D**. In the present embodiment, the transmitting filters **7A** and **7B** can work concurrently for simultaneous transmission of signals. The relationship between the conductor provided for the transmitting filter **7A** and the conductor provided for the transmitting filter **7D** applies to the relationship between the conductor provided for the transmitting filter **7A** and the conductor provided for the transmitting filter **7B**. The transmitting filters **7A** and **7C** can also work concurrently for simultaneous transmission of signals. The relationship between the conductor provided for the transmitting filter **7A** and the conductor provided for the transmitting filter **7D** also applies to the relationship between the conductor provided for the transmitting filter **7A** and the conductor provided for the transmitting filter **7C**. This yields an improvement in the heat dissipation properties of any two or more transmitting filters capable of working concurrently for simultaneous transmission of signals and thus eliminates or reduces the possibility that the characteristics of each of the transmitting filters **7A** to **7D** will deteriorate.

(97) (3) Principal Effects

(98) The radio-frequency module **1** according to the embodiment described above includes the mounting substrate **30** (substrate), the transmitting filters (e.g., the transmitting filters **7A** and **7D**, a resin layer **32A**, the shield layer **33**, the conductors (e.g., the conductors **401** and **404**). The mounting substrate **30** has the first main surface **35** and the second main surface **36** on opposite sides. The transmitting filter **7A** is mounted on the first main surface **35** of the mounting substrate **30** and is a filter whose passband is the first frequency band. The transmitting filter **7D** is mounted on the first main surface **35** of the mounting substrate **30** and is a filter whose passband is the second frequency band. The resin layer **32A** is disposed on the first main surface **35** of the mounting substrate **30**. The transmitting filters **7A** and **7D** are at least partially covered with the resin layer **32A**. The shield layer **33** is disposed on the outer surface of the resin layer **32A**. The conductor **401** is disposed on the first main surface **35** of the mounting substrate **30** and is in contact with the transmitting filter **7A** and the mounting substrate **30**. The conductor **404** is disposed on the first main surface **35** of the mounting substrate **30** and is in contact with the transmitting filter **7D** and the mounting substrate **30**. The transmitting filters **7A** and **7D** are used to enable simultaneous transmission. The conductor **401** is in contact with the shield layer **33** on a side other than the side closer to the transmitting filter **7D** (the second filter) than to the transmitting filter **7A** (the first filter). The conductor **404** is in contact with the shield layer **33** on a side other than the side closer to the transmitting filter **7D** (the first filter) than to the transmitting filter **7A** (the second filter). This feature reduces the possibility that heat generated by the transmitting filter **7A** will cause deterioration of the characteristics of the transmitting filter **7D** while both the transmitting filters **7A** and **7D** are used for simultaneous transmission. This feature also reduces the possibility that heat generated by the transmitting filter **7D** will cause deterioration of the characteristics of the transmitting filter **7A**.

(99) (4) Modifications

(100) The following describes modifications of the aforementioned embodiment.

(101) (4-1) Modification 1

(102) This modification is an example that involves changing the direction in which the conductors **401** and **404** in the aforementioned embodiment extend. When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**) as illustrated in FIG. 7, the conductor **401** provided for the transmitting filter **7A** in this modification extends from the transmitting filter **7A** to one side in a direction (the third direction **D3**) orthogonal to the direction (the second direction **D2**) in which the transmitting filters **7A** and **7D** are aligned. With the shield layer **33** having side surfaces respectively denoted by **332** to **335**, the conductor **401** is in contact with the side surface **334**, which is located on the side to which the conductor **401** extends.

(103) When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**), the conductor **404** provided for the transmitting filter **7D** in this modification extends from the transmitting filter **7D** to the other side in the direction (the third direction **D3**) orthogonal to the direction (the second direction **D2**) in which the transmitting filters **7A** and **7D** are aligned; that is, the conductors **401** and **404** extend to opposite sides in the direction concerned. The conductor **404** is in contact with one of the four side surfaces of the shield layer **33** that are respectively denoted by **332** to **335**. More specifically, the conductor **404** is in contact with the side surface **335** located on the side to which the conductor **404** extends. Alternatively, the conductors **401** and **404** may extend to the same side in the direction concerned and both may be in contact with the side surface **334**. As in the embodiment described above, the conductor **404** provided for the transmitting filter **7D** in this modification also extends from the transmitting filter **7D** to the side surface **333** on the front side of the shield layer **33** and is in contact with the side surface **333**.

(104) More specifically, the conductor **404** includes two constituent conductors, which are denoted by **404a** and **404b**, respectively. The two constituent conductors respectively denoted by **404a** and **404b** are connected to each other on the first main surface **35** of the mounting substrate **30** or, more

specifically, in the region in which the transmitting filter 7D (the transmitting filter for which the conductor **404** is provided) is placed, that is, in the region onto which the shape of the transmitting filter 7D is projected. The constituent conductor **404a** extends from the transmitting filter 7D (the transmitting filter for which the conductor **404** is provided) to the side surface **333** on the front side of the shield layer **33** and is in contact with the side surface **333** on the front side. When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction D1), the constituent conductor **404b** extends from the transmitting filter 7D (the transmitting filter for which the conductor **404** is provided) to one side (e.g., the right side) in the direction orthogonal to the constituent conductor **404a** and is in contact with the side surface **335** on the right side of the shield layer **33**. Alternatively, the conductor **401** and the constituent conductor **404b** may extend to the same side in the direction concerned and both may be in contact with the side surface **334**. The two constituent conductors respectively denoted by **404a** and **404b** are in contact with the respective side surfaces. With the shield layer **33** having surface portions respectively denoted by **331** to **335**, the constituent conductors **404a** and **404b** are in contact with the side surfaces **333** and **335**, respectively.

(105) With the shield layer **33** having surface portions respectively denoted by **331** to **335**, the conductor **404** is in contact with two or more of the surface portions of the shield layer **33**. For example, the conductor **404** is in contact with the side surfaces **333** and **335** as illustrated in FIG. 7. This feature yields a further improvement in the heat dissipation properties of the transmitting filter 7D.

(106) With a further improvement in the heat dissipation properties of the transmitting filter 7D, this modification further reduces the possibility that heat generated by the transmitting filter 7D will cause deterioration of the characteristics of the transmitting filter 7A.

(107) (4-2) Modification 2

(108) This modification is an example that involves changing the arrangement of the conductors **401** and **404** in the aforementioned embodiment and the direction in which the conductors **401** and **404** extend. In this modification (see FIG. 8), the two transmitting filters (the transmitting filters 7A and 7D) capable of working concurrently for simultaneous transmission of signals are located in the midsection of the first main surface **35** of the mounting substrate **30** and are arranged side by side in the front-and-rear direction (i.e., in the second direction D2).

(109) The conductor **401** provided for the transmitting filter 7A extends from the transmitting filter 7A to the side surface **332** located on the upper side of the shield layer **33** and is in contact with the side surface **332** on the upper side.

(110) The conductor **404** provided for the transmitting filter 7D extends from the transmitting filter 7D to the three side surfaces of the shield layer **33** that are respectively denoted by **332** to **335** and is in contact with the side surfaces **332** to **335**. More specifically, the conductor **404** includes three constituent conductors, which are denoted by **404a**, **404b**, and **404c**, respectively. The three constituent conductors respectively denoted by **404a**, **404b**, and **404c** are connected to each other on the first main surface **35** of the mounting substrate **30** or, more specifically, in the region in which the transmitting filter 7D (the transmitting filter for which the conductor **404** is provided) is placed, that is, in the region onto which the shape of the transmitting filter 7D is projected.

(111) The constituent conductor **404a** extends from the transmitting filter 7D (the transmitting filter for which the conductor **404** is provided) to the side surface **333** on the front side of the shield layer **33** and is in contact with the side surface **333** on the front side. When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction D1), the constituent conductors **404b** and **404c** extend from the transmitting filter 7D (the transmitting filter for which the conductor **404** is provided) to the respective sides in the (left-and-right) direction (i.e., the third direction D3) orthogonal to the constituent conductor **404a** and is in contact with the side surface **334** on the left side of the shield layer **33** and the side surface **335** on the right side of the shield layer **33**, respectively. That is, the conductor **404** is in contact with two or more surface portions.

For example, the conductor **404** is in contact with three side surfaces respectively denoted by **333**, **334**, and **335** as illustrated in FIG. **8**. This feature yields a further improvement in the heat dissipation properties of the transmitting filter **7D**. This modification further reduces the possibility that heat generated by the transmitting filter **7D** will cause deterioration of the characteristics of the transmitting filter **7A**.

(112) (4-3) Modification 3

(113) This modification is an example that involves changing the arrangement of the conductors **401** and **404** in the aforementioned embodiment and the direction in which the conductors **401** and **404** extend. In this modification (see FIG. **9**), one of the two transmitting filters (the transmitting filters **7A** and **7D**) capable of working concurrently for simultaneous transmission of signals, namely, the transmitting filter **7D** is located at a corner **35s** of the first main surface **35** of the mounting substrate **30**. To be more specific, the transmitting filter **7D** is close to the corner **35s**. The other transmitting filter, namely, the transmitting filter **7A** is disposed on a desired part of the first main surface **35** of the mounting substrate **30**. For example, the transmitting filter **7A** is close to the middle of the first main surface **35**.

(114) As in Modification 2, the conductor **401** provided for the transmitting filter **7A** extends from transmitting filter **7A** to the side surface **332** located on the rear side of the shield layer **33** and is in contact with the side surface **332** on the rear side.

(115) The conductor **404** provided for the transmitting filter **7D** extends from the transmitting filter **7D** to two side surfaces of the shield layer **33** respectively denoted by **333** and **335** and is in contact with the side surfaces **332** and **335**. More specifically, the conductor **404** includes two constituent conductors and an extension. The two constituent conductors are denoted by **404a** and **404b**, respectively. The extension is denoted by **404d**.

(116) The two constituent conductors respectively denoted by **404a** and **404b** are connected to each other on the first main surface **35** of the mounting substrate **30** or, more specifically, in the region in which the transmitting filter **7D** (the transmitting filter for which the conductor **404** is provided) is placed, that is, in the region onto which the shape of the transmitting filter **7D** is projected. The constituent conductor **404a** extends from the transmitting filter **7D** (the transmitting filter for which the conductor **404** is provided) to the side surface **333** on the front side of the shield layer **33** and is in contact with the side surface **333** on the front side. When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**), the constituent conductor **404b** extends from the transmitting filter **7D** (the transmitting filter for which the conductor **404** is provided) to one side (e.g., the right side) in the direction orthogonal to the constituent conductor **404a** and is in contact with the side surface **335** on the right side of the shield layer **33**. With the shield layer **33** having four side surfaces respectively denoted by **332** to **335**, the side surfaces **333** and **335** are two side surfaces that meet at the corner **35s**.

(117) The extension **404d** is L-shaped at the corner **35s** and is connected to tip portions of the two constituent conductors respectively denoted by **404a** and **404b** on the first main surface **35** of the mounting substrate **30**. With the shield layer **33** having four side surfaces respectively denoted by **332** to **335**, the extension **404d** is in contact with the side surfaces **333** and **335**, namely, the two side surfaces that meet at the corner **35s**. When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**), the extension **404d** (the conductor **404**) extends along two side surfaces respectively denoted by **333** and **335**.

(118) More specifically, the extension **404d** includes a first extension **50** and a second extension **51**. The first extension **50** extends along the side surface **333** on the front side of the shield layer **33** and is in contact with the side surface **333**; that is, the first extension **50** extends in the left-and-right direction of the mounting substrate **30** (i.e., in the third direction **D3**). Referring to FIG. **9**, the first extension **50** extends across the mounting substrate **30** in the third direction **D3**; that is, the first extension **50** extends from the corner **35s** to an end portion opposite to the corner **35s**. The second extension **51** extends along the side surface **335** on the right side of the shield layer **33** and is in

contact with the side surface **335**; that is, the second extension **51** extends in the front-and-rear direction of the mounting substrate **30** (i.e., in the second direction **D2**). Referring to FIG. **9**, the second extension **51** extends in the second direction **D2** from the corner **35s** to about an end portion **7a** of the transmitting filter **7D**. The end portion **7a** of the mounting substrate **30** is one of two end portions of the mounting substrate **30** that are located on opposite sides in the second direction **D2**, and the end portion **7a** is farther than the other end portion from the corner **35s**.

(119) The conductor **404** in this modification provides the following effect: with the shield layer **33** having side surfaces respectively denoted by **332** to **335**, the transmitting filter **7D** corresponding to the conductor **404** and disposed at the corner **35s** of the first main surface **35** of the mounting substrate **30** is effectively in contact with two side surfaces that meet at a corner (e.g., the side surfaces **333** and **335** that meet at the corner **35s**). This feature yields a further improvement in the heat dissipation properties of the transmitting filter **7D**. This modification further reduces the possibility that heat generated by the transmitting filter **7D** will cause deterioration of the characteristics of the transmitting filter **7A**.

(120) (4-4) Modification 4

(121) This modification concerns an example in which three transmitting filters (e.g., the transmitting filters **7A** to **7C**) in the aforementioned embodiment can work concurrently for simultaneous transmission of signals. The conductors provided for the transmitting filters **7A** to **7C** are denoted by **401** to **403**, respectively. In this modification (see FIG. **10**), the three transmitting filters (the transmitting filters **7A**, **7B**, and **7C**) on the first main surface **35** of the mounting substrate **30** are located in the midsection in the third direction **D3** and are arranged side by side in the second direction **D2**.

(122) The conductor **401** provided for the transmitting filter **7A** is in contact with the shield layer **33** on a side other than the side closer to the transmitting filters **7B** and **7C** than to the transmitting filter **7A**. The conductor **402** provided for the transmitting filter **7B** is in contact with the shield layer **33** on a side other than the sides closer respectively to the transmitting filters **7A** and **7C** than to the transmitting filter **7B**. The conductor **403** provided for the transmitting filter **7C** is in contact with the shield layer **33** on a side other than the side closer to the transmitting filters **7A** and **7B** than to the transmitting filter **7C**.

(123) Referring to FIG. **10**, the conductor **401** provided for the transmitting filter **7A** extends from the transmitting filter **7A** to the side surface **332** located on the upper side of the shield layer **33** and is in contact with the side surface **332** on the upper side. The conductor **402** provided for the transmitting filter **7B** extends from the transmitting filter **7B** to the side surface **335** located on the right side of the shield layer **33** and is in contact with the side surface **335** on the right side. The conductor **403** provided for the transmitting filter **7C** extends from the transmitting filter **7C** to the side surface **333** located on the lower side of the shield layer **33** and is in contact with the side surface **333** on the lower side.

(124) Effects similar to those of the aforementioned embodiment are produced accordingly; that is, this modification yields an improvement in the heat dissipation properties of each of the three transmitting filters respectively denoted by **7A** to **7C** and capable of working concurrently for simultaneous transmission of signals and thus eliminates or reduces the possibility that the characteristics of the transmitting filters **7A** to **7C** will deteriorate.

(125) (4-5) Modification 5

(126) This modification concerns an example in which the conductor **404** provided for the transmitting filter **7D** in the aforementioned embodiment is in contact with the top surface **331** of the shield layer **33**. Referring to FIG. **11A** and FIG. **11B**, the radio-frequency module **1** in this modification includes a partition member **56**, which is electrically conductive. For example, the partition member **56** is a compartment shield. More specifically, the partition member **56** is disposed on a desired part of the first main surface **35** of the mounting substrate **30** to partition the inner part of the radio-frequency module **1** (i.e., the part surrounded with the mounting substrate **30**

and the shield layer 33). The partition member 56 is made of an electrically conductive material (e.g., metal). The partition member 56 on the first main surface 35 of the mounting substrate 30 is in the form of a wall. The partition member 56 on the first main surface 35 of the mounting substrate 30 is on the right side of the transmitting filters 7A and 7D and extends in the second direction D2 (see FIG. 11A). The partition member 56 has a top surface 56a, which is in contact with the back side of the top surface 331 of the shield layer 33 (see FIG. 11B).

(127) The conductor 404 in this modification includes a conductor main body 49 and the partition member 56. The conductor main body 49 is disposed on the first main surface 35 of the mounting substrate 30 and is in contact with the partition member 56 and the transmitting filter 7D for which the conductor 404 is provided. More specifically, the conductor main body 49 is, for example, strip-shaped with its contact region extending along the first main surface 35 of the mounting substrate 30. As described above, the conductors 401 and 404 in the aforementioned embodiment are in contact with the transmitting filter 7A and 7D, respectively. An end portion of the conductor main body 49 in this modification is in contact with the corresponding transmitting filter in the same fashion as the conductors 401 and 404 in the aforementioned embodiment; that is, the end portion of the conductor main body 49 lies in the region in which the transmitting filter 7D is placed, and the end portion is in contact with the support substrate 71 of the transmitting filter 7D with the connection member 41 (e.g., a solder bump) therebetween. The other end portion of the conductor main body 49 is in contact with the partition member 56. Referring to FIG. 10, the other end portion of the conductor main body 49 is located between the partition member 56 and the mounting substrate 30. That is, the already-existing partition member 56 is available for use as part of the conductor 404.

(128) With the conductor 404 given as above, the heat Q2 generated by the transmitting filter 7D is transferred to the support substrate 71, the connection member 41, the conductor main body 49, and the partition member 56 in sequence. The heat is then transferred from the top surface 56a of the partition member 56 to the top surface 331 of the shield layer 33 and is dissipated from the shield layer 33.

(129) This modification provides the following effects. Although a top surface 7S of the transmitting filter 7D is not in contact with the back side of the top surface 331 of the shield layer 33, the heat Q2 generated by the transmitting filter 7D is dissipated through the top surface 331 of the shield layer 33. With the addition of the conductor main body 49 to the already-existing partition member 56, that is, with such structural simplicity, the heat Q2 generated by the transmitting filter 7D is dissipated through the top surface 331 of the shield layer 33.

(130) (4-6) Modification 6

(131) The conductor 404 in Modification 5 makes use of the partition member 56 in the form of a wall, whereas the conductor 404 (a first conductor) in this modification makes use of a partition member 55, which has an internal accommodation part 555 defined therein. More specifically, the partition member 55 in this modification (see FIG. 12) is made of an electrically conductive material (e.g., copper) and is a box-shaped rectangular parallelepiped. The partition member 55 has a space (the internal accommodation part 555) in which a circuit board 53 populated with electronic components can be placed. That is, the partition member 55 in this modification divides the inner part of the radio-frequency module 1 (i.e., the part surrounded with the mounting substrate 30 and the shield layer 33) into an inner side and an outer side. The partition member 55 is mounted on the first main surface 35 of the mounting substrate 30. The partition member 55 has a top surface 55a, which is in contact with the back side of the top surface 331 of the shield layer 33.

(132) More specifically, the partition member 55 includes a substrate 551, a shield wall 552, and connection members 553. For example, the shield wall 552 is made of an electrically conductive material (e.g., copper) and defines a box-shaped rectangular parallelepiped whose bottom is open. With the shield wall 552 being disposed on the substrate 551, the open bottom of the parallelepiped

defined by the shield wall **552** is covered with the substrate **551**. The space defined by the substrate **551** and the shield wall **552** is the internal accommodation part **555**, in which the circuit board **53** is placed. The circuit board **53** is mounted on the substrate **551**. The connection members **553** each forms an electrical connection between the substrate **551** and the mounting substrate **30**. For example, the connection member **553** is a solder bump. At least one of the connection members **553** is a connection member for establishing a ground. The partition member **55** in the present embodiment defines an internal hollow space. Alternatively, a resin layer may be fitted in the space. The shield wall **552** has a top surface **552a**, which is also regarded as the top surface **55a** of the partition member **55** and is in contact with the back side of the top surface **331** of the shield layer **33**.

(133) In this modification, the conductor **404** provided for the transmitting filter **7D** includes a conductor main body **49** and the partition member **55**. The conductor main body **49** is similar to the conductor main body **49** in Modification 5. That is, the conductor main body **49** is, for example, strip-shaped with its contact region extending along the first main surface **35** of the mounting substrate **30**. As in the aforementioned embodiment, an end portion of the conductor main body **49** is in contact with the transmitting filter **7D** with the connection member **41** (e.g., a solder bump) therebetween. The other end portion of the conductor main body **49** on the first main surface **35** of the mounting substrate **30** lies in the region surrounded by the partition member **55** and is in contact with the back surface of the partition member **55** (i.e., the back surface of the substrate **551**) with a connection member **556** (e.g., a solder bump) therebetween. The substrate **551** includes a wiring conductor, and the connection member **556** may be in contact with the shield wall **552** with the wiring conductor therebetween. This means that the conductor main body **49** may be in contact with the shield wall **552** with the connection member **556** and the wiring conductor of the substrate **551** therebetween. That is, as with the partition member in Modification 5, the already-existing partition member **55** in this modification is available for use as part of the conductor **404**.

(134) With the conductor **404** given as above, the heat **Q2** generated by the transmitting filter **7D** is transferred to the support substrate **71**, the connection member **41**, the conductor main body **49**, and the partition member **55** in sequence, just as in the case of Modification 5. The heat is then transferred from the top surface **55a** of the partition member **55** to the top surface **331** of the shield layer **33** and is dissipated from the top surface **331**.

(135) As with Modification 5, this modification provides the following effects. Although the top surface of the transmitting filter **7D** is not in contact with the top surface **331** of the shield layer **33**, the heat generated by the transmitting filter **7D** is dissipated through the top surface **331** of the shield layer **33**. With the addition of the conductor main body **49** to the already-existing partition member **55**, that is, with such structural simplicity, the heat generated by the transmitting filter **7D** is dissipated through the top surface **331** of the shield layer **33**.

(136) In this modification, a conductor **54** (a second conductor) may be disposed at the bottom of the internal accommodation part **555** of the partition member **55** (i.e., on a front surface **551a** of the substrate **551**). The conductor **54** is a member that forms an electrical connection between the circuit board **53** and the shield wall **552**. The circuit board **53** is in contact with the shield wall **552** with the conductor **54** therebetween. Heat generated by the circuit board **53** is denoted by **Q3**. With the conductor **54** given as above, the heat **Q3** is transferred through the conductor **54** and the shield wall **552** to the top surface **331** of the shield layer **33** and is dissipated from the top surface **331**. That is, the partition member **55** serves as both a heat transfer path for the transmitting filter **7D** and a heat transfer path for the circuit board **53**. Thus, the heat generated by the circuit board **53** placed in the internal space defined by the partition member **55** is dissipated through the conductor **54**, the partition member **55**, and the shield layer **33**.

(137) (4-7) Modification 7

(138) This modification is illustrated in FIGS. **13A** and **13B**, in which signal lines disposed in the mounting substrate **30** overlap conductors serving as heat transfer paths for filters. For example,

signal lines respectively denoted by **81** and **82** overlap conductors denoted respectively by **402** and **501**. The conductors **402** and **501** each serves as a heat transfer path for the transmitting filter **7B** and a heat transfer path for the receiving filter **8A**. Not all of the transmitting filters **7A** to **7D** and the receiving filters **8A** to **8D** are illustrated in FIGS. **13A** and **13B**. More specifically, the transmitting filters **7A** and **7B** and the receiving filter **8A** are illustrated in FIGS. **13A** and **13B**. (139) As illustrated FIG. **13B**, the mounting substrate **30** in this modification is, for example, a multilayer substrate. The transmitting filters **7A** and **7B**, the receiving filter **8A**, and the conductors **401**, **402**, and **501** are mounted on the first main surface **35** of the mounting substrate **30**. The conductors **401**, **402**, and **501** are provided for the transmitting filter **7A**, the transmitting filter **7B**, and the receiving filter **8A**, respectively.

(140) The conductors **401**, **402**, and **501** are disposed on the first main surface **35** of the mounting substrate **30** (see FIG. **13A**). One end portion of the conductor **401** lies in the region in which the transmitting filter **7A** (the transmitting filter for which the conductor **401** is provided) is placed. The end portion of the conductor **401** is in contact with the transmitting filter **7A** with the connection member **41** therebetween. The other end portion of the conductor **401** extends from the transmitting filter **7A** to a side (the left side) other than the side closer to the transmitting filter **7B** and the receiving filter **8A** than to the transmitting filter **7A** and is in contact with the side surface **334** of the shield layer **33**. One end portion of the conductor **402** lies in the region in which the transmitting filter **7B** (the transmitting filter for which the conductor **402** is provided) is placed. The end portion of the conductor **402** is in contact with the transmitting filter **7B** with the connection member **41** therebetween. The other end portion of the conductor **402** extends from the transmitting filter **7B** to a side (the rear side) other than the sides closer respectively to the transmitting filter **7A** and the receiving filter **8A** than to the transmitting filter **7B** and is in contact with the side surface **332** of the shield layer **33**. One end portion of the conductor **501** lies in the region in which the receiving filter **8A** (the transmitting filter for which the conductor **501** is provided) is placed. The end portion of the conductor **501** is in contact with the receiving filter **8A** with the connection member **41** therebetween. The other end portion of the conductor **501** extends from the receiving filter **8A** to a side (the rear side) other than the side closer to the transmitting filters **7A** and **7B** than to the receiving filter **8A** and is in contact with the side surface **332** of the shield layer **33**.

(141) The first switch **14** is disposed on the second main surface **36** of the mounting substrate **30**. Signal lines respectively denoted by **81** to **83** are disposed in the mounting substrate **30**. Transmission signals or reception signals may be transmitted through the signal lines **81** to **83**. For example, the signal line **81** forms an electrical connection between the transmitting filter **7A** and the selection terminal **14h** of the first switch **14**, and the signal line **82** forms an electrical connection between the transmitting filter **7B** and the selection terminal **14i** of the first switch **14**. The signal line **83** forms an electrical connection between the receiving filter **8A** and the selection terminal **14d** of the first switch **14**.

(142) The signal line **81** extends from the transmitting filter **7A** and passes through the inside of the mounting substrate **30**. The signal line **81** extends below the conductor **402** and is connected to the selection terminal **14h** of the first switch **14**. The signal line **82** extends from the transmitting filter **7B** and passes through the inside of the mounting substrate **30**. The signal line **82** extends below the conductor **403** and is connected to the selection terminal **14i** of the first switch **14**. The signal line **83** extends from the receiving filter **8A** and passes through the inside of the mounting substrate **30**. The signal line **83** is connected to the selection terminal **14d** of the first switch **14**. It is not required that the signal lines **81** to **83** be disposed in the mounting substrate **30**; that is, it is required that the signal lines **81** to **83** be disposed in the mounting substrate **30** and/or on the second main surface **36** of the mounting substrate **30**.

(143) When viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**) as illustrated in FIG. **13A** and FIG. **13B**, the signal line **81** in this modification at

least partially overlaps the conductor **402**. The signal line **81** may at least partially overlap the conductor **501** when viewed in plan in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**). Likewise, the signal line **82** may at least partially overlap the conductor **501**. (144) The signal line **81** and the conductor **402** in this modification overlap each other when viewed in the thickness direction of the mounting substrate **30** (i.e., in the first direction **D1**). The conductor **402** eliminates or reduces the possibility that noise (an electromagnetic wave) produced in the signal line **81** will come into the transmitting filter **7B**. The conductor **402** also eliminates or reduces the possibility that noise produced in the transmitting filter **7B** will come into the signal line **81**. The same holds true for the signal line **82** and the conductor **501**, which overlap each other. The conductor **501** eliminates or reduces the possibility that noise produced in the signal line **82** will come into the receiving filter **8A**. The conductor **501** also eliminates or reduces the possibility that noise produced in the receiving filter **8A** will come into the signal line **82**. Thus, the signal line **81** and the transmitting filter **7B** are electromagnetically shielded from each other. Likewise, the signal line **82** and the receiving filter **8A** are electromagnetically shielded from each other.

(145) (4-8) Other Modifications

(146) An embodiment has been described in which the transmitting filters **7A** and **7D** work concurrently for simultaneous transmission of signals. In some embodiments, however, any two or more of the receiving filters **8A** to **8D** work concurrently for simultaneous reception of signal, in which case conductors such as the conductors **401** and **404** in the aforementioned embodiment may serve as heat transfer paths for the two or more of the receiving filters **8A** to **8D**.

(147) In the aforementioned embodiment, at least one of the transmitting filters **7A** to **7D** and at least one of the receiving filters **8A** to **8D** may work concurrently (for simultaneous transmission and reception of signals). The present disclosure is applicable to such an embodiment, in which case the at least one of the transmitting filters **7A** to **7D** and at least one of the receiving filters **8A** to **8D** are used in combination.

(148) The mounting substrate **30** in the aforementioned embodiment is in the form of a rectangular plate. In some embodiments, the mounting substrate **30** is generally in the form of a polygonal plate or a circular plate. The shield layer **33** and the mounting substrate **30** has the same outside shape when viewed in plan. For example, both the outer shape of the mounting substrate **30** and the outer shape of the shield layer **33** are polygonal or circular when viewed in plan.

(149) The aforementioned embodiment and Modifications 1 to 7 may be employed in combination.

(150) In the aforementioned embodiment, two transmitting filters or more specifically, the transmitting filters **7A** and **7D** are capable of working concurrently for simultaneous transmission of signals and are connected to the power amplifiers **5A** and **5B**, respectively. In this case, the maximum power output of the power amplifier **5A** may be greater than the maximum output of the power amplifier **5B**. Only one of the two transmitting filters capable of working concurrently may be provided with a conductor. More specifically, the transmitting filter **7A** connected to the power amplifier **5A** whose maximum power output is greater than the maximum power output of the power amplifier **5B** may be provided with the conductor **401**; that is, the conductor **404** for the transmitting filter **7D** connected to the power amplifier **5B** may be eliminated. Settings may be made to ensure that the heat dissipation performance of the conductor **401** provided for the transmitting filter **7A** connected to the power amplifier **5A** whose maximum power output is greater than the maximum power output of the power amplifier **5B** is higher than the heat dissipation performance of the conductor **404** provided for the transmitting filter **7D** connected to the power amplifier **5B**.

(151) When the maximum power output of the power amplifier **5A** is greater than the maximum power output of the power amplifier **5B**, the amount of heat generated by the power amplifier **5A** is greater than the amount of heat generated by the power amplifier **5B**. When the power amplifier **5A** is running hot, the heat generated by the power amplifier **5A** can get into the power amplifier **5B**. As a result, the characteristics of the power amplifier **5B** can deteriorate. The same goes for the

transmitting filters 7A and 7D that are connected to the power amplifiers 5A and 5B, respectively. That is, the amount of heat generated by the transmitting filter 7A connected to the power amplifier 5A whose maximum power output is greater than the maximum power output of the power amplifier 5B is greater than the amount of heat generated by the transmitting filter 7D connected to the power amplifier 5B. For this reason, the transmitting filter that can run hot, namely, the transmitting filter 7A is provided with the conductor 401. With an improvement in the heat dissipation properties of the transmitting filter 7A, this modification reduces the possibility that heat generated by the transmitting filter 7A will cause deterioration of the characteristics of the transmitting filter 7D.

(152) As mentioned above, this modification is made with a focus on the maximum power output of each of the power amplifiers 5A and 5B. Alternatively, modifications may be made with a focus on power classes to which the power amplifiers 5A and 5B belong. The term power classes are categories of terminals and are based on their power output determined in accordance with, for example, the maximum power output. A power class with a smaller numeral denotes that the terminals in the power class concerned are capable of providing higher power. The maximum power output of devices in a high power class is greater than the maximum power output of devices in a non-high power class. For example, a method specified by 3GPP is used to measure the maximum power output. As for the aforementioned case, the power amplifier 5A belongs to the high power class, whereas the power amplifier 5B belongs to the non-high power class. Only the transmitting filter 7A connected to the power amplifier 5A in the smaller power class may be provided with the conductor 401; that is, the conductor 404 for the transmitting filter 7D connected to the power amplifier 5B in the smaller power class may be eliminated.

(153) (5) Aspects of Disclosure

(154) The following aspects of the present disclosure are disclosed herein.

(155) In a first aspect of the present disclosure, a radio-frequency module (1) includes a mounting substrate (30), a first filter (e.g., 7A), a second filter (e.g., 7D), a resin layer (32A), a shield layer (33), and a conductor (e.g., 401). The mounting substrate (30) has a first main surface (35) and a second main surface (36) on opposite sides. The first filter (7A) is mounted on the first main surface (35) of the mounting substrate (30). The first filter (7A) is a filter whose passband is a first frequency band. The second filter (7D) is mounted on the first main surface (35) of the mounting substrate (30). The second filter (7D) is a filter whose passband is a second frequency band. The resin layer (32A) is disposed on the first main surface (35) of the mounting substrate (30). The first filter (7A) and the second filter (7D) are at least partially covered with the resin layer (32A). The shield layer (33) is disposed on an outer surface of the resin layer (32A). The radio-frequency module (1) is capable of performing simultaneous transmission or simultaneous reception by using both the first filter (7A) and the second filter (7D). The conductor (401) is disposed on the first main surface (35) of the mounting substrate (30) and is in contact with the first filter (7A) and the mounting substrate (30). The conductor (401) is in contact with the shield layer (33) on a side other than a side closer to the second filter (7D) than to the first filter (7A).

(156) The conductor (e.g., 401) provided as above reduces the possibility that heat generated by the first filter (e.g., 7A) will cause deterioration of the characteristics of the second filter (e.g., 7D) while both the first filter (7A) and the second filter (7D) are used for simultaneous transmission. More specifically, the conductor (401) serving as a heat transfer path for the first filter (7A) is less susceptible to heat generated by the second filter (7D). The resultant improvement in the heat dissipation properties of the first filter (7A) eliminates or reduces the possibility that heat generated by the first filter (7A) will affect the second filter (7D). This reduces the possibility that the heat generated by the first filter (7A) will cause deterioration of the characteristics of the second filter (7D). The conductor (401) serving as a heat transfer path for the first filter (7A) eliminates or reduces the possibility that heat transferred through the heat transfer path for the first filter (7A) will affect the second filter (7D). The resultant improvement in the heat dissipation properties of

the second filter (7D) eliminates or reduces the possibility that heat generated by the second filter (7D) will affect the first filter (7A). This reduces the possibility that the heat generated by the second filter (7D) will cause deterioration of the characteristics of the first filter (7A).

(157) In a second aspect, the radio-frequency module (1) according to the first aspect further includes a first power amplifier (5A) and a second power amplifier (5B). The first power amplifier (5A) is connected to the first filter (7A). The second power amplifier (5B) is connected to the second filter (7D).

(158) With the addition of the first power amplifier (5A) and the second power amplifier (5B) that are connected respectively to the first filter (7A) and the second filter (7D), the aforementioned effect will be produced; that is, the present disclosure eliminates or reduces the possibility that the characteristics of each of the first filter (7A) and the second filter (7D) will deteriorate.

(159) In a third aspect, the radio-frequency module (1) according to the second aspect is as follows: the maximum power output of the first power amplifier (e.g., 7A) is greater than the maximum power output of the second power amplifier (e.g., 7D).

(160) The conductor (404) enables an improvement in the heat dissipation properties of the first filter (7A) connected to the first power amplifier (5A), namely, the power amplifier whose maximum power output is greater than the other power amplifier.

(161) In a fourth aspect, the radio-frequency module (1) in any one of the first to third aspects is as follows. The first filter (e.g., 7A) is an acoustic wave filter. The first filter (7A) includes an IDT electrode (71c) and a support substrate (71). The support substrate (71) supports the IDT electrode (71c). The first filter (e.g., 7A) is mounted on the mounting substrate (30) with the support substrate (71) being located between the mounting substrate (30) and the IDT electrode (71c).

(162) When heat generated by the first filter (e.g., 7A) arises from the support substrate (71), this placement enables an effective heat transfer from the support substrate (71) to the conductor (401). This feature yields a further improvement in the heat dissipation properties of the first filter (e.g., 7A).

(163) In a fifth aspect, the radio-frequency module (1) according to any one of the first to fourth aspects is as follows. The shield layer (33) has a plurality of surface portions (331 to 335) with which the outer surface of the resin layer (32A) is covered. The conductor (401) is in contact with one of the plurality of surface portions (331 to 335) other than another one of the plurality of surface portions (331 to 335) that is closer to the second filter (e.g., 7D) than to the first filter (e.g., 7A).

(164) When the shield layer (33) has the plurality of surface portions (331 to 335) (i.e., when the first main surface (35) of the mounting substrate (30) has a polygonal shape), the aforementioned effect will be produced; that is, the present disclosure eliminates or reduces the possibility that the characteristics of each of the first filter (7A) and the second filter (7D) will deteriorate.

(165) In a sixth aspect, the radio-frequency module (1) according to the fifth aspect is as follows. The plurality of surface portions (331 to 335) include a plurality of side surfaces (332 to 335) of the shield layer (33). The conductor (e.g., 401) is in contact with one of the plurality of side surfaces (332 to 335) that is closer than the other ones of the plurality of side surfaces to the first filter (e.g., 7A).

(166) This feature enables the minimization of the length of the conductor (e.g., 401) serving as the heat dissipation path for the first filter (e.g., 7A). The heat dissipation properties of the first filter (e.g., 7A) will be improved accordingly.

(167) In a seventh aspect, the radio-frequency module (1) according to the fifth or sixth aspect is as follows: the conductor (e.g., 401) is in contact with two or more of the plurality of surface portions (331 to 335) of the shield layer (33).

(168) This feature yields an improvement in the heat dissipation properties of the first filter (e.g., 7A).

(169) In an eighth aspect, the radio-frequency module (1) according to any one of the fifth to

seventh aspects is as follows. The plurality of surface portions (331 to 335) include a plurality of side surfaces (332 to 335) of the shield layer (33). The first main surface (35) of the mounting substrate (30) has a corner (35s). The first filter (e.g., 7D) is disposed at the corner (35s) of the first main surface (35) of the mounting substrate (30). The conductor (e.g., 404) extends along the corner (35s) of the first main surface (35) of the mounting substrate (30) and is in contact with two of the plurality of side surfaces (332 to 335) of the shield layer (33). The two side surfaces (e.g., 333 and 335) meet at the corner (35s).

(170) The conductor (e.g., 404) given as above provides the following effect: with the shield layer (33) having the side surfaces (332 to 335), the first filter (e.g., 7D) disposed at the corner (35s) of the first main surface (35) of the mounting substrate (30) is effectively in contact with the two side surfaces (e.g., 333 and 335) that meet at the corner (35s). The heat dissipation properties of the first filter (e.g., 7D) will be improved accordingly.

(171) In a ninth aspect, the radio-frequency module (1) according to the eighth aspect is as follows: the conductor (404) extends along the two side surfaces (e.g., 333 and 335) when viewed in plan in a thickness direction (D1) of the mounting substrate (30).

(172) The conductor (404) given as above has a contact region extending along the outline (e.g., the L-shape) of the corner (35s).

(173) In a tenth aspect, the radio-frequency module (1) according to any one of the first to ninth aspects is as follows. A signal line (e.g., 81) through which a transmission signal or a reception signal flows is disposed in the mounting substrate (30) and/or on the second main surface (36) of the mounting substrate (30). The signal line (e.g., 81) at least partially overlaps the conductor (e.g., 402) when viewed in plan in a thickness direction (D1) of the mounting substrate (30).

(174) With this placement, the first filter (e.g., 7B) and the signal line (e.g., 81) are electromagnetically shielded from each other by the conductor (e.g., 402). More specifically, the conductor (402) eliminates or reduces the possibility that electromagnetic noise produced in the signal line (81) will come into the conductor (402) (i.e., into the first filter (7B)). The conductor (402) also eliminates or reduces the possibility that electromagnetic noise produced in the first filter (7B) will come into the signal line (81).

(175) In an eleventh aspect, the radio-frequency module (1) according to any one of the first to tenth aspects is as follows. The conductor (404) includes a conductor main body (49) and a partition member (56 or 55) that is electrically conductive. The conductor main body (49) is disposed on the first main surface (35) of the mounting substrate (30) and is in contact with the first filter (7A). The partition member (56 or 55) is disposed on the first main surface (35) of the mounting substrate (30) and is in contact with the conductor main body (49) and the shield layer (33).

(176) This feature provides the following effects. Although a surface (e.g., a top surface) of the first filter (e.g., 7D) is not in contact with a surface (e.g., a top surface (331)) of the shield layer (33), heat generated by the first filter (7D) is dissipated through the surface (e.g., the top surface (331)) of the shield layer (33). With the addition of the conductor main body (49) to the already-existing partition member (56 or 55), that is, with such structural simplicity, the heat generated by the first filter (7D) is dissipated through the surface (e.g., the top surface (331)) of the shield layer (33).

(177) In a twelfth aspect, the radio-frequency module (1) according to the eleventh aspect is as follows. The conductor (404) is a first conductor (404). The partition member (55) has an internal accommodation part (555) in which a circuit board (53) populated with electronic components is placed. The circuit board (53) is in contact with the partition member (55) with a second conductor (54) being disposed between the circuit board (53) and the partition member (55) and being disposed on a bottom surface of the partition member (55) (i.e., a front surface (551a) of a substrate (551) of the partition member (55)).

(178) This feature provides the following effect: heat generated by the circuit board (53) placed in the internal space defined by the partition member (55) is dissipated through the second conductor

(54), the partition member (55) and is then dissipated through the top surface (331) of the shield layer (33).

(179) In a thirteenth aspect, the radio-frequency module (1) according to any one of the first to twelfth aspects is as follows. The conductor (401) is a first conductor (401). The radio-frequency module (1) further includes a third filter (7C), a second conductor (402), and a third conductor (403). The third filter (7C) is mounted on the first main surface (35) of the mounting substrate (30). The third filter (7C) is a filter whose passband is a third frequency band. The second conductor (402) is disposed on the first main surface (35) of the mounting substrate (30) and is in contact with the second filter (7B) and the mounting substrate (30). The third conductor (403) is mounted on the first main surface (35) of the mounting substrate (30) and is in contact with the third filter (7C) and the mounting substrate (30). The first filter (7A), the second filter (7B), and the third filter (7C) are each at least partially covered with the resin layer (32A). The radio-frequency module (1) is capable of performing simultaneous transmission by using the first filter (7A), the second filter (7B), and the third filter (7C). The first conductor (401) is in contact with the shield layer (33) on a side other than a side closer to the second filter (7B) and the third filter (7C) than to the first filter (7A). The second conductor (402) is in contact with the shield layer (33) on a side other than sides closer respectively to the first filter (7A) and the third filter (7C) than to the second filter (7B). The third conductor (403) is in contact with the shield layer (33) on a side other than a side closer to the first filter (7A) and the second filter (7B) than to the third filter (7C).

(180) With this feature, the present disclosure is applicable to the first to third filters (7A to 7C) that are used for simultaneous transmission. This yields an improvement in the thermal dissipation properties of each of the first to third filters (7A to 7C) that are used for simultaneous transmission.

(181) In a fourteenth aspect, a communication device (100) includes the radio-frequency module (1) according to any one of the first to thirteenth aspects and a signal processing circuit (2). The signal processing circuit (2) is connected to the radio-frequency module (1) and is configured to process a radio-frequency signal.

(182) With this configuration, the communication device (100) including the radio-frequency module (1) is capable of producing the aforementioned effects.

REFERENCE SIGNS LIST

(183) 1 radio-frequency module 2 signal processing circuit 3A to 3C antenna 5A power amplifier (first power amplifier) 5B power amplifier (second power amplifier) 6A to 6D low-noise amplifier 7A transmitting filter (first filter, second filter) 7B transmitting filter (first filter, second filter) 7C transmitting filter (third filter) 7D transmitting filter (first filter, second filter) 8A to 8B receiving filter 10A, 10B output matching circuit 11A to 11D, 12A to 12C matching circuit 14 first switch 14a to 14c common terminal 14d to 14k selection terminal 15 second switch 15a common terminal 15b to 15d selection terminal 16 third switch 16a to 16c common terminal 16d to 16g selection terminal 17 controller 18A to 19j external connection terminal 21 baseband signal processing circuit 22 RF signal processing circuit 30 mounting substrate (first substrate) 31 circuit component 32A first resin layer 32B second resin layer 33 shield layer 35 first main surface 35L first half region 35R second half region 35s corner 36 second main surface 401 conductor (first conductor) 402 conductor (second conductor) 403 conductor (third conductor) 404 conductor (first conductor) 501 conductor 41 connection member 49 conductor main body 50 first extension 51 second extension 53 circuit board (substrate) 54 conductor (second conductor) 55, 56 partition member 55a, 56a top surface 71 support substrate 71a, 71b main surface 71c IDT electrode 71d pad electrode 72 spacer layer 73 cover member 74 through-via electrode 75 connection terminal 81 to 83 signal line 100 communication device 331 top surface 331 to 335 surface portion 332 to 335 side surface 404a to 404c constituent conductor 404d extension 551 substrate 551a front surface 552 shield wall 552a top surface 553 connection member 555 internal accommodation part 556 connection member D1 first direction (thickness direction) D2 second direction D3 third direction Q1 to Q3 heat R1 to R4, T1 to T4 signal path Tc common path T21, T31, T41 individual path

Claims

1. A radio-frequency module, comprising: a mounting substrate comprising a first main surface and a second main surface on opposite sides; a first filter on the first main surface of the mounting substrate, the first filter comprising a filter including a passband that comprises a first frequency band; a second filter on the first main surface of the mounting substrate, the second filter comprises a filter including a passband that comprises a second frequency band; a resin layer on the first main surface of the mounting substrate, the first and second filters being at least partially covered with the resin layer; a shield layer on an outer surface of the resin layer; and a conductor on the first main surface of the mounting substrate and being in contact with the first filter and the mounting substrate, wherein the radio-frequency module is configured to perform simultaneous transmission that utilizes both the first filter and the second filter, and the conductor is in contact with the shield layer on a side different than a side closer to the second filter than to the first filter.
2. The radio-frequency module according to claim 1, further comprising: a first power amplifier connected to the first filter; and a second power amplifier connected to the second filter.
3. The radio-frequency module according to claim 2, wherein a first maximum power output of the first power amplifier is greater than a second maximum power output of the second power amplifier.
4. The radio-frequency module according to claim 3, wherein the first filter comprises an acoustic wave filter, the first filter includes an interdigital transducer (IDT) electrode and a support substrate that is configured to support the IDT electrode, and the first filter is on the mounting substrate with the support substrate between the mounting substrate and the IDT electrode.
5. The radio-frequency module according to claim 3, wherein the shield layer comprises a plurality of surface portions with which the outer surface of the resin layer is covered, and the conductor is in contact with a first of the plurality of surface portions other than a second of the plurality of surface portions that is closer to the second filter than to the first filter.
6. The radio-frequency module according to claim 2, wherein the first filter comprises an acoustic wave filter, the first filter includes an interdigital transducer (IDT) electrode and a support substrate that is configured to support the IDT electrode, and the first filter is on the mounting substrate with the support substrate between the mounting substrate and the IDT electrode.
7. The radio-frequency module according to claim 2, wherein the shield layer comprises a plurality of surface portions with which the outer surface of the resin layer is covered, and the conductor is in contact with a first of the plurality of surface portions other than a second of the plurality of surface portions that is closer to the second filter than to the first filter.
8. The radio-frequency module according to claim 7, wherein the plurality of surface portions comprise a plurality of side surfaces of the shield layer, and the conductor is in contact with one of the plurality of side surfaces that is closer than the other ones of the plurality of side surfaces to the first filter.
9. The radio-frequency module according to claim 1, wherein the first filter comprises an acoustic wave filter, the first filter includes an interdigital transducer (IDT) electrode and a support substrate that is configured to support the IDT electrode, and the first filter is on the mounting substrate with the support substrate between the mounting substrate and the IDT electrode.
10. The radio-frequency module according to claim 9, wherein the shield layer comprises a plurality of surface portions with which the outer surface of the resin layer is covered, and the conductor is in contact with a first of the plurality of surface portions other than a second of the plurality of surface portions that is closer to the second filter than to the first filter.
11. The radio-frequency module according to claim 1, wherein the shield layer comprises a plurality of surface portions with which the outer surface of the resin layer is covered, and the conductor is in contact with a first of the plurality of surface portions other than a second of the

plurality of surface portions that is closer to the second filter than to the first filter.

12. The radio-frequency module according to claim 11, wherein the plurality of surface portions comprise a plurality of side surfaces of the shield layer, and the conductor is in contact with one of the plurality of side surfaces that is closer than the other ones of the plurality of side surfaces to the first filter.

13. The radio-frequency module according to claim 11, wherein the conductor is in contact with two or more of the plurality of surface portions of the shield layer.

14. The radio-frequency module according to claim 11, wherein the plurality of surface portions comprise a plurality of side surfaces of the shield layer, the first main surface of the mounting substrate comprises a corner, the first filter is at the corner of the first main surface of the mounting substrate, and the conductor extends along the corner of the first main surface of the mounting substrate and is in contact with two of the plurality of side surfaces of the shield layer, with the two side surfaces meeting at the corner.

15. The radio-frequency module according to claim 14, wherein the conductor extends along the two side surfaces when viewed in plan in a thickness direction of the mounting substrate.

16. The radio-frequency module according to claim 1, wherein a signal line, through which a transmission signal or a reception signal flows, is in the mounting substrate and/or on the second main surface of the mounting substrate, and the signal line at least partially overlaps the conductor when viewed in plan in a thickness direction of the mounting substrate.

17. The radio-frequency module according to claim 1, wherein the conductor comprises: a conductor main body on the first main surface of the mounting substrate and being in contact with the first filter, and a partition member on the first main surface of the mounting substrate and being in contact with the conductor main body and the shield layer, the partition member being electrically conductive.

18. The radio-frequency module according to claim 17, wherein the conductor comprises a first conductor, the partition member comprises an internal accommodation part that comprises a circuit board that includes one or more electronic components, and the circuit board is in contact with the partition member with a second conductor being between the circuit board and the second conductor and being on a bottom surface of the partition member.

19. The radio-frequency module according to claim 1, wherein the conductor comprises a first conductor, the radio-frequency module further comprises: a third filter on the first main surface of the mounting substrate, the third filter comprising a filter including a passband that comprises a third frequency band, a second conductor on the first main surface of the mounting substrate and being in contact with the second filter and the mounting substrate, and a third conductor on the first main surface of the mounting substrate and being in contact with the third filter and the mounting substrate, the first filter, the second filter, and the third filter are each at least partially covered with the resin layer, the radio-frequency module is further configured to perform simultaneous transmission by utilizing the first filter, the second filter, and the third filter, the first conductor is in contact with the shield layer on a side different than a side closer to the second filter and the third filter than to the first filter, the second conductor is in contact with the shield layer on a side different than sides closer respectively to the first filter and the third filter than to the second filter, and the third conductor is in contact with the shield layer on a side different than a side closer to the first filter and the second filter than to the third filter.

20. A communication device, comprising: the radio-frequency module according to claim 1; and a signal processing circuit connected to the radio-frequency module and configured to process a radio-frequency signal.
